Propagation of Export Shocks: The Great Recession in Japan*

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Abstract

During the Great Recession, Japanese exports collapsed by 36.1%. Japan's real GDP fell by 8.8%, nearly three times the U.S. decline, despite minimal direct exposure to the U.S. housing market. We quantify the propagation of the export shock by developing a multi-sector, multi-region real business cycle model calibrated using Japan's inter-regional input-output tables, with region-industry export shocks measured from customs data. We find that the measured export shocks account for about three-fourths of the observed GDP decline; consumption demand linkages are quantitatively as important as intermediate-input linkages; propagation follows a gravity pattern across regions; and first-order approximations markedly understate propagation.

Keywords: Great Recession, export demand, inter-regional input-output table, multi-sector model

JEL Classification: D57, E32, F41, F44, R15

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1 Introduction

The Great Recession, which spanned 2007 to 2009, began with the collapse of the U.S. housing market. It was not only the largest U.S. post-war recession up to that point, but also had a global impact. Japan's real GDP fell by 8.8% between the first quarter of 2008 and the first quarter of 2009. This decline exceeded the 3.3% fall in U.S. real GDP over the same period, even though Japanese financial institutions had little direct exposure to the U.S. housing market. Japanese exports declined even more sharply: their real value fell by 36.1% during this period. Given this collapse in exports, a large share of which went to the U.S., it is natural to deduce that some of the decline in Japanese GDP was caused by a reduction in export demand, which arrived in Japan as an exogenous shock.[1]

This event is a rare natural experiment in which (i) macroeconomic shocks arrive in a large economy with little anticipation; (ii) shocks ("the impulse") are identifiable at the regional and industry levels using customs-level trade data; and (iii) the regional and sectoral links ("the propagation") can be traced using inter-regional input-output (IRIO) tables. Thus, a detailed examination of this event provides important insight into the macroeconomic propagation mechanism of exogenous shocks. The Japanese government constructs detailed IRIO tables every five years, which are an essential element of our analysis. Such tables are not common—the U.S., for example, does not have such comprehensive information on region-industry-level linkages.

We analyze the effect of this large export decline in Japan using a multi-sector, multi-region business cycle model. Shocks to exports from a particular region and industry propagate to other regions and industries through two channels. First, reduced demand in one region and industry lowers demand for intermediate goods from another region and industry. Second, the decline in production reduces consumers' income through lower wages and profits. The resulting decline in consumption demand in another region also serves as a propagation mechanism.

We ask two questions. The first is: to what extent does this single shock account for the massive decline in Japanese GDP in this episode? Although the sharp decline in Japanese GDP and exports during the Great Recession has been documented descriptively (e.g., Sommer, 2009), the extent to which the negative export demand shock translated into a decline in domestic demand, and the channels through which it did so, have not been systematically explored. Addressing this question requires a structural model with detailed data on exports and domestic transactions to trace the propagation of the export shock. Second, how does a foreign demand shock that arrives in a particular region affect outcomes in other regions and industries? In other words, how does a shock propagate across regions and industries? This question is central to business cycle theory but has been difficult to answer because observable exogenous shocks and their propagation channels are scarce. Our setting makes it possible to answer this question directly.

Using a calibrated quantitative model, we find that the measured export shocks account for about three-fourths of the GDP decline from 2008Q3 to 2009Q1 and almost all of the consumption

[1] We treat the export demand shock as an exogenous shock from the viewpoint of the Japanese economy. Bems et al. (2010) show that the large decline in durable demand was a major cause of the 2008-2009 trade collapse. A more recent paper by Miyamoto and Nguyen (2024) finds a similar result. Eaton et al. (2016) estimate a multicountry general equilibrium model and argue that the trade collapse during the Great Recession is mainly caused by the shift in spending away from tradable sectors. Some other studies examine the supply-side factors. For example, Amiti and Weinstein (2011) analyze how trade finance affected the decline in exports and find that trade finance can explain less than half of the export decline in Japan during this period. Therefore, treating the export decline observed in the data as primarily caused by a decline in exogenous foreign demand is a reasonable approximation.

decline. The model explains regional declines well in regions with high export exposure (Chūbu, Chūgoku, Kyūshū). The export shocks generate substantial declines even in regions like Tōhoku with small export shares, indicating substantial inter-regional propagation.

The counterfactual decompositions yield three findings. First, consumption demand linkages are quantitatively as important as intermediate-input linkages. This finding complements the existing network literature, which primarily emphasizes input-output linkages. Second, consistent with the international trade literature, shock propagation follows a gravity pattern. Importantly, the strength of this gravity pattern varies across channels: it is pronounced for investment demand but essentially absent for consumption demand, implying that regional propagation decays with distance for investment demand but not for consumption demand. This heterogeneity has not been documented in the literature. Finally, price adjustment substantially attenuates shock propagation.

Our model exhibits strong nonlinearity: first-order approximations substantially understate the effects of large export shocks. In an experiment with a shock of similar magnitude to the observed one—a 50% decline in export demand for a particular region-industry—the first-order approximation yields a decline in national sales that is only one-third as large as that predicted by the full model.

Our study advances the real business cycle research program, in the tradition of Kydland and Prescott (1982) and Long and Plosser (1983). Since Frisch (1933), macroeconomists have analyzed the business cycle through the lens of shocks and their propagation. Since the 1980s, this approach has given rise to both vector autoregressions and the real business cycle approach. Skeptics of real business cycle theory, such as Summers (1986) in response to Prescott (1986), have criticized the difficulty of interpreting the "shocks," particularly the technology shocks that were dominant in the early contributions. Cochrane (1994) also emphasizes this difficulty. We consider shocks that are particularly relevant to the Japanese economy during the Great Recession: export demand shocks.[2] The progress we make here is that we can identify the "impulse" in two dimensions (exports from particular regions and industries) and trace the propagation process through (i) the IRIO matrix and (ii) production, consumption, and labor supply decisions in the model.

The use of a foreign demand shock, rather than a productivity or policy shock, as an impulse is also an important and distinctive feature of our study. Using a simplified version of the model, we clarify why a foreign demand shock induces comovement in GDP, consumption, and labor input: the change in the terms of trade affects the economy much like a productivity shock. Beyond this particular episode, a foreign demand shock is an important source of instability, particularly for a small economy, and our analysis sheds light on the mechanism. In this context, a closely related paper is Gorodnichenko et al. (2012), which quantifies the contribution of the collapse in Soviet demand and rising energy costs to Finland's recession using a multi-sector general equilibrium model. Our paper differs in two important respects. First, we explicitly model domestic input-output linkages across industries and regions, whereas their model abstracts from direct input-output linkages among its three domestic sectors. Second, their analysis combines the loss of the Soviet export market with an effective increase in imported energy costs. In contrast, we isolate the effects of foreign demand shocks without imposing another shock. Our monopolistically competitive framework models demand shocks for differentiated exports and allows us to quantify

[2] Mendoza (1995) is an early contribution incorporating exogenous international trade shocks (in the form of a "terms-of-trade shock" in his case) into a small open economy real business cycle framework. His model takes the terms of trade as given, whereas export prices are determined endogenously in our model. His three-sector model abstracts from domestic regional linkages, whereas we analyze propagation across both industries and regions.

how endogenous price adjustment affects their propagation.

Many recent studies examine how shocks propagate and amplify through the IO network (Atalay, 2017; Vom Lehn and Winberry, 2022; Liu and Tsyvinski, 2024). Classic contributions include Long and Plosser (1983), Dupor (1999), Horvath (2000), Foerster et al. (2011), and Acemoglu et al. (2012). This literature primarily focuses on how productivity shocks in one industry propagate across industries and affect the aggregate economy. This study advances the literature by examining how demand shocks propagate across regions. In particular, our quantitative and analytical results speak to the nonlinear effects of the production network (Baqaee and Farhi, 2019; Dew-Becker, 2023), regional multiplier effects (Nakamura and Steinsson, 2014; Dupor et al., 2023; Flynn et al., 2024), and a gravity model of inter-regional transactions in tradable intermediate goods (Allen and Arkolakis, 2025).

Some recent papers also study the propagation of trade shocks. Huneeus (2020) analyzes the Chilean economy during the Great Recession using a firm-to-firm production network model. Dhyne et al. (2022) consider the Belgian economy. As in our analysis, both papers treat the export shock as an exogenous decline in foreign demand. Their focus is on the firm-to-firm network, whereas the current paper highlights propagation across industries and regions. Aggregating the production network to the industry level ensures the unique existence of equilibrium, which is not guaranteed in a general network. An emerging literature examines the global propagation of shocks through the international input-output network. Examples include Huo et al. (2023), Ho et al. (2024), and Boeckelmann et al. (2024). Our paper is analogous in that we emphasize comovement across regions through the input-output network. However, these papers cannot analyze a demand shock from outside the set of regions they model because their global-economy models have no "outside." Our model is unique in that the demand shock comes from outside the set of regions, and we can measure the impulse using customs data.

A study closely related to our paper is Caliendo et al. (2018), who analyze the propagation of regional and sectoral productivity changes in the U.S. economy. Our study differs from theirs in four respects. First, their static, perfectly competitive framework builds on Eaton and Kortum (2002) and abstracts from international trade, whereas ours is a dynamic small open economy model featuring monopolistic competition. We conduct counterfactual experiments to quantify the role of price adjustment in shock propagation. Second, we study foreign demand shocks, with magnitudes calibrated to the large export declines observed during the Great Recession. Third, their model treats services as nontradable across regions: interregional trade flows are measured using the Commodity Flow Survey for manufacturing industries. Our inter-regional input-output data capture trade in both goods and services, allowing us to quantify propagation through interregional service linkages. Recent studies emphasize that service trade is significant even in the international context (Han et al., 2025). The tradability of services is expected to be higher within a country. Finally, their model allows labor to move freely across regions, whereas we assume labor is immobile across regions. The low annual interregional migration rate, approximately 1% of the population across our nine regions in 2008, supports this short-run approximation.

The remainder of this paper is organized as follows. Section 2 reviews key macro facts about the Great Recession in Japan. Section 3 describes the model. Section 4 presents analytical results in a special case to illustrate the basic mechanism underlying export-demand-led output fluctuations. Section 5 computes the model and calibrates it to the data. Section 6 simulates the model using the export series. Section 7 conducts counterfactual experiments and decomposes the channels through which regional GDP changes. Section 8 presents a simplified model to examine propagation factors

and analyzes the model's nonlinearity. Section 9 concludes.

2 Overview of the Great Recession in Japan

In this section, we present time-series patterns of various statistics from Japan during the Great Recession. We present statistics for the country as a whole, by region, and by industry. Our data primarily come from public sources, including the Ministry of Economy, Trade and Industry (METI), the Research Institute of Economy, Trade and Industry (RIETI), the Ministry of Health, Labour and Welfare (MHLW), the Ministry of Finance (MoF), and the Statistics Bureau of Japan. METI has provided inter-regional input-output tables (IRIO) until 2005, constructed for 9 regions and x industries, where $x \in \{12, 29, 53\}$.[3] We reconstruct an IRIO with 26 industries and 9 regions using the 2005 tables so that the industry classification aligns with the Prefectural Account.[4] We use the JIP database from the RIETI to compute the technology parameter later. The Labour Force Survey (LFS) from the Statistics Bureau and the Monthly Labour Survey (MLS) conducted by the MHLW are used to calibrate parameters related to labor supply for each region.[5] Finally, to construct regional and sectoral export data, we use the Trade Statistics of Japan (TSJ), compiled by the MoF.

2.1 Time series of GDP and export

Figure 1 plots the quarterly real GDP time series (seasonally adjusted, in billions of 2015 yen), with annual values. One can see the sharp drop in 2008. From the first quarter of 2008 to the first quarter of 2009, real GDP declined by approximately 8.8%. Figure 2 plots the time series of exports of goods and services (seasonally adjusted, in billions of 2015 yen).[6] The decline in exports from the first quarter of 2008 to the first quarter of 2009 was larger than that of GDP, amounting to 36.1%.

Figure 3 plots the export time series by industry. All industries saw exports decline during the Great Recession, with the decline particularly sharp for the transportation equipment (TE) industry. The TE industry includes the automobile industry, and this outcome largely reflects the decline in automobile demand in the U.S.

2.2 Regional heterogeneity

A key aspect of our analysis is the explicit consideration of regional linkages. Figure 4 shows how we divide Japan into nine regions: Hokkaidō, Tōhoku, Kantō, Chūbu, Kansai,[7] Chūgoku, Shikoku,

[3] The construction method of IRIO is detailed in Arai (2020) and references therein. Transactions in manufacturing sectors are based on the Commodity Distribution Survey. Interregional transactions in services are reported when estimation is feasible. Some service sectors, such as construction, are categorically considered non-tradable. Factors considered include margins in the retail and financial sectors, service consumption by tourists and commuters, and back-office and managerial services provided internally within multi-regional firms.

[4] The Prefectural Account is the prefectural version of the GDP statistics. See, https://www.esri.cao.go.jp/jp/sna/sonota/kenmin/kenmin_top.html (only the Japanese version is available).

[5] See, https://www.stat.go.jp/english/index.html.

[6] Both series are taken from https://www.esri.cao.go.jp/jp/sna/data/data_list/sokuhou/files/2022/qe221_2/tables/gaku-jk2212.csv.

[7] Kansai corresponds to "Kinki" in the inter-regional input-output dataset in Japanese. We follow METI's English expression for the region.

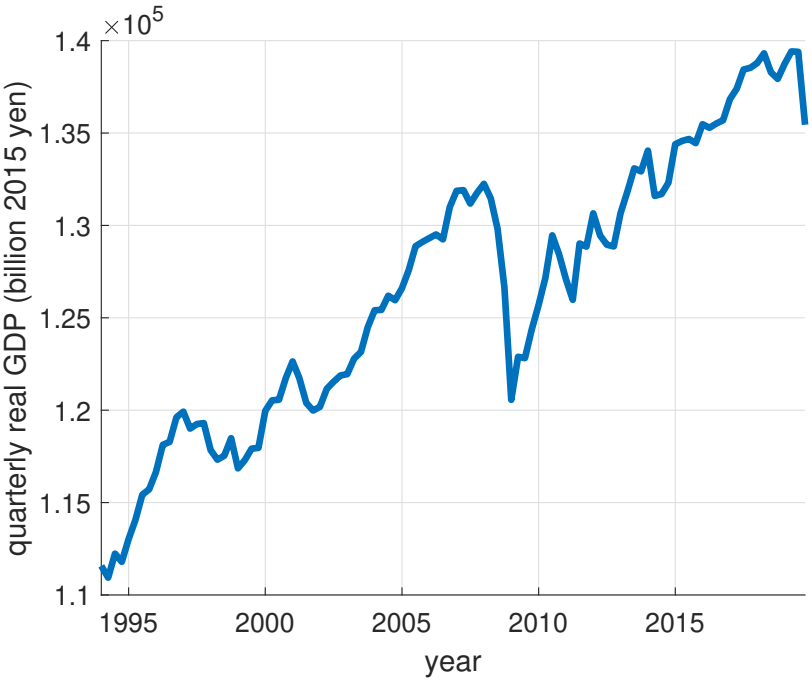


Figure 1: Time series of Japanese GDP

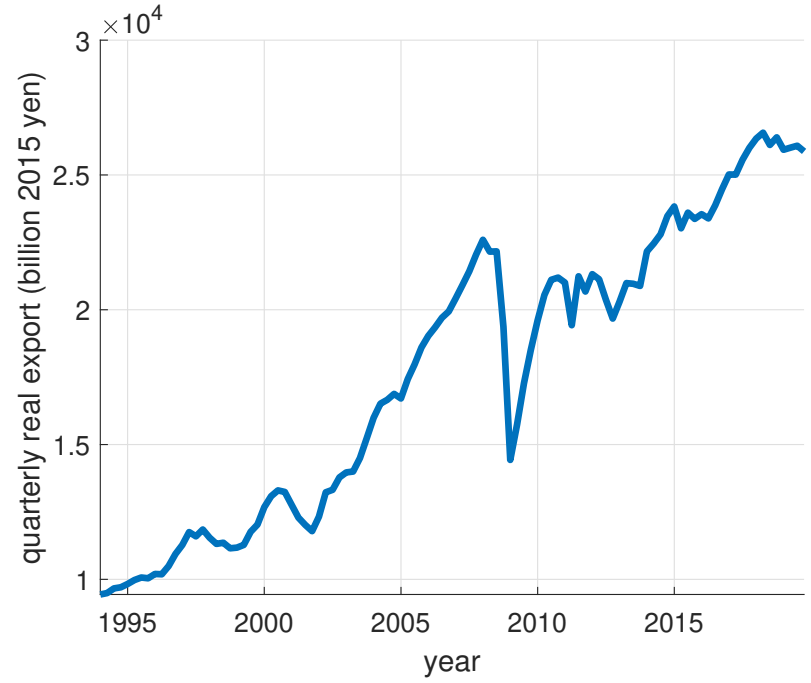


Figure 2: Time series of Japanese exports

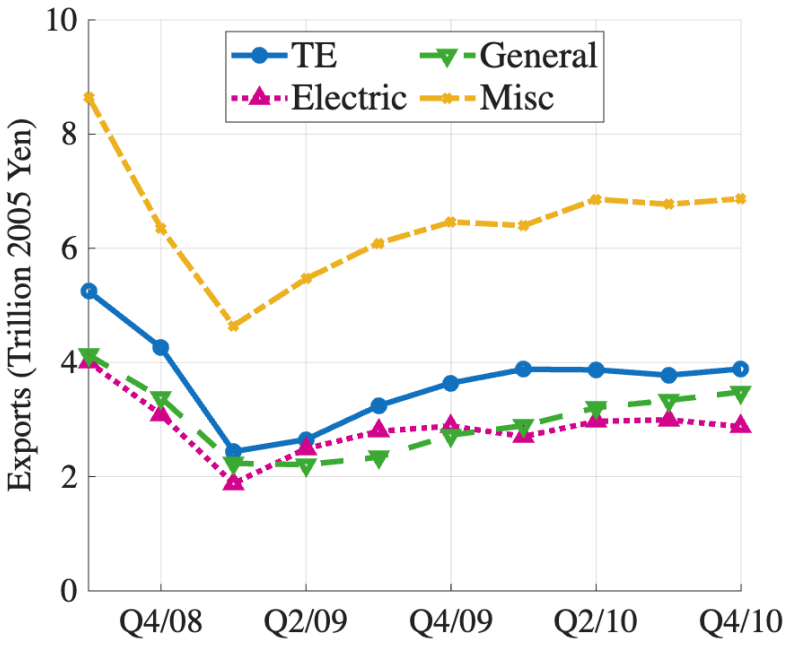


Figure 3: Real sectoral exports: 2008Q3−2010Q4. "TE," "General," "Electric," and "Misc" refer to Transportation Equipment, General Machinery, Electric Machinery, and Miscellaneous (comprising the rest of the 23 industries).

Kyūshū, and Okinawa. This division is primarily motivated by the availability of inter-regional IO matrices. The precise mapping of prefectures to regions is provided in Appendix A. In Figure 4, darker colors indicate higher regional GDP. Kantō, including the Tōkyō area, is the largest economic region among the nine. Kansai includes the Ōsaka area, the second-largest economic region, and Chūbu includes Nagoya, the third-largest economic region. Chūbu also hosts the headquarters of Toyota, the largest automaker and auto exporter.

Figure 5 plots the time series of regional real GDP. All regions except Okinawa experienced a significant decline in GDP during the Great Recession. We also observe considerable heterogeneity across regions in the magnitude of the decline.

Figure 6 shows the export series for each region from 2008Q3 to 2010Q4, expressed as a share of 2008Q3 GDP in that region. The construction of the export data, drawn from the TSJ, is detailed in Appendix B. The time series reveals considerable heterogeneity across regions, both in industry composition and in the magnitude of the decline in exports during the Great Recession. The most severe shock occurred in the transportation equipment sector in the Chūbu region: its export value declined by 62.8% in the first quarter of 2009.

2.3 Gravity effects in inter-regional transactions

Because the IRIO data include a geographical dimension, the tables reveal a gravity structure of inter-regional trade in intermediate demands for manufacturing and services. This section investigates the quantitative impact of distance on inter-regional transactions. This exercise is not only of independent interest, as empirical studies on inter-regional gravity structures are scarce, but also

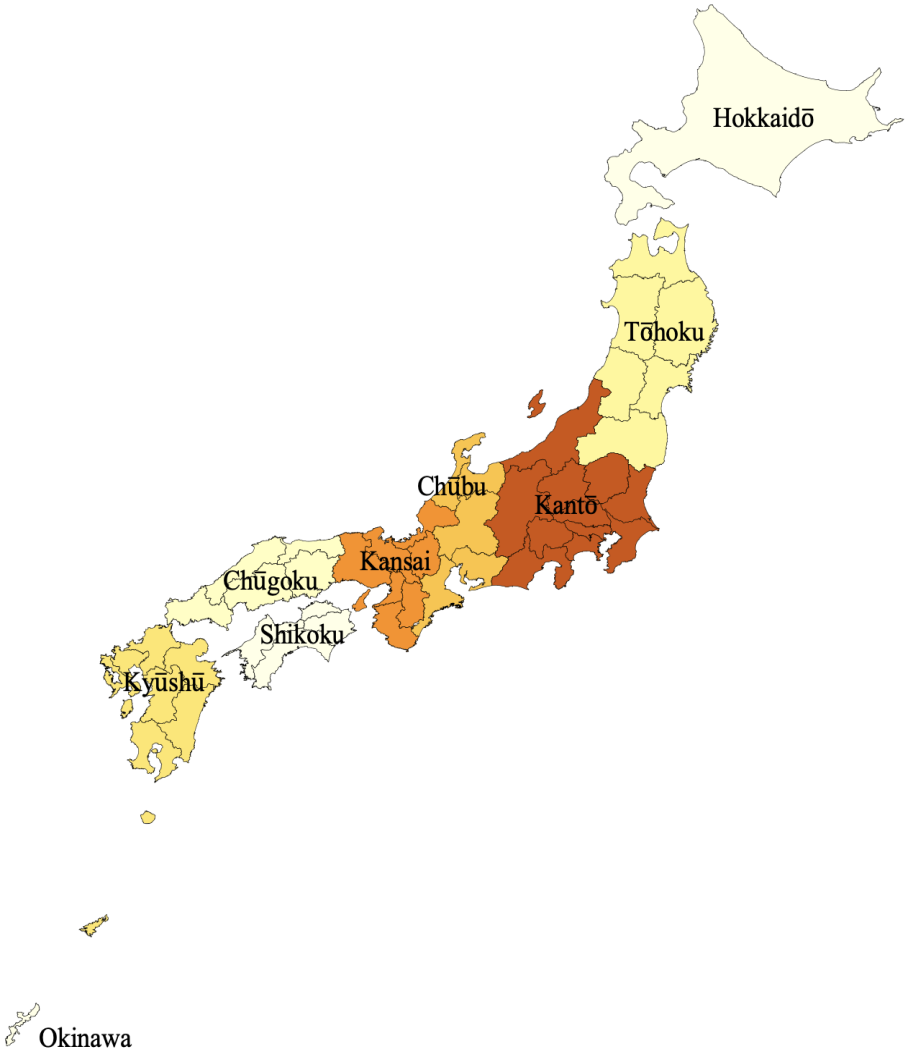


Figure 4: Division of Japanese Regions

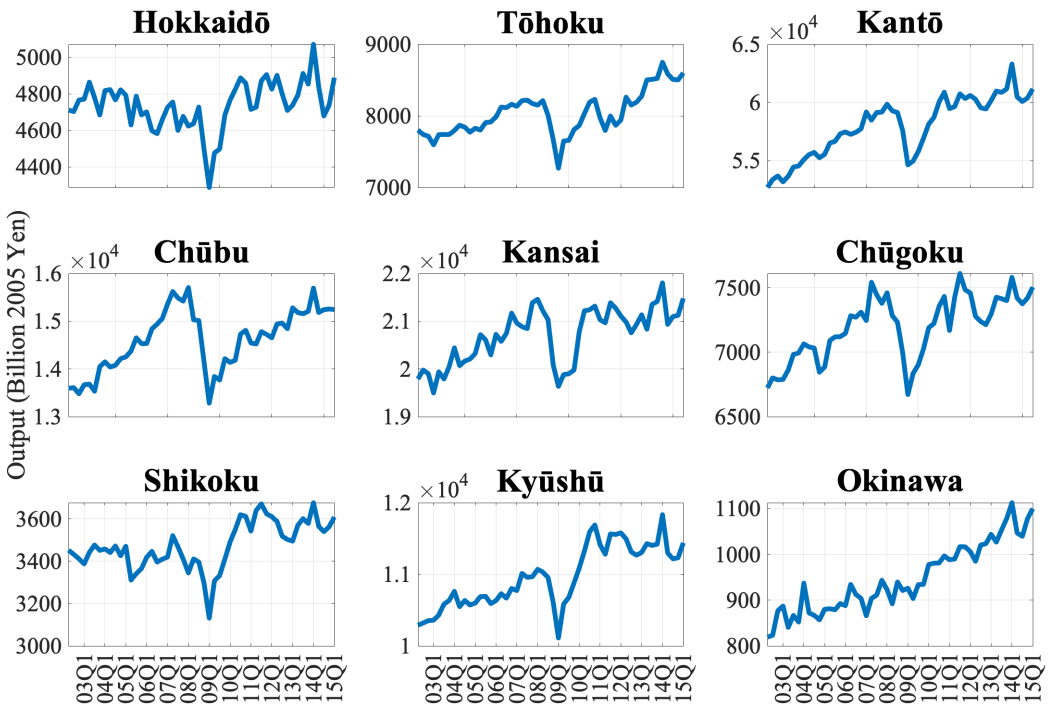


Figure 5: Regional GDP: 2002Q4−2015Q1

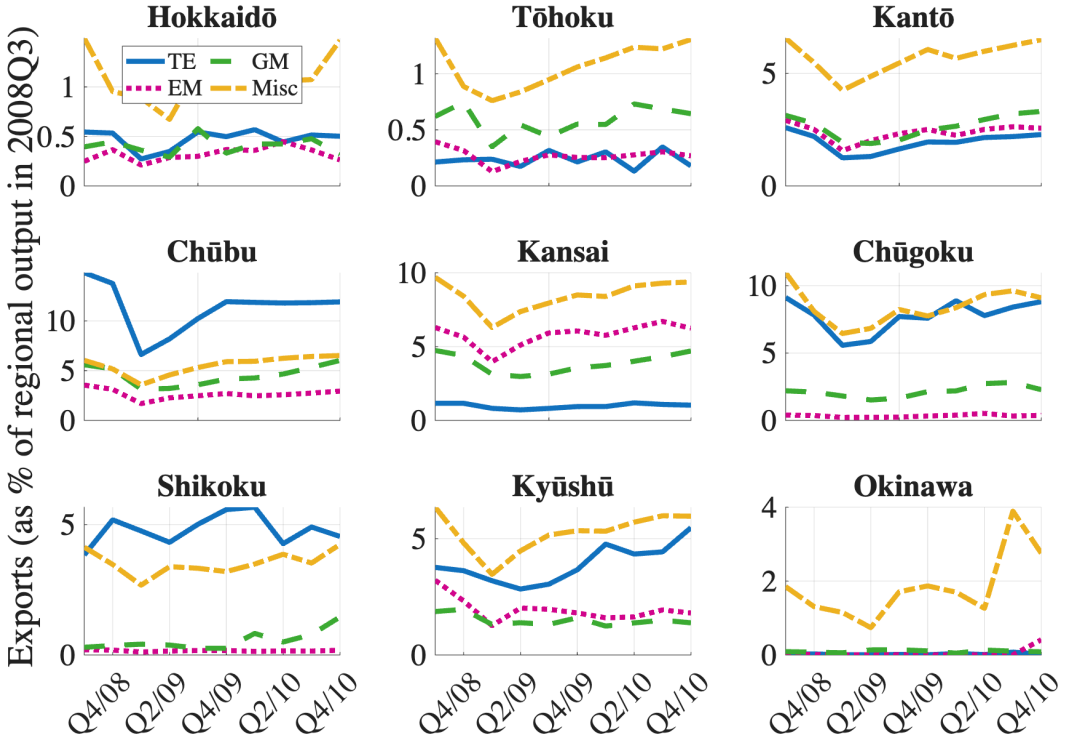


Figure 6: Regional exports: 2008Q3−2010Q4. "TE", "GM", "EM", and "Misc" refer to Transportation Equipment, General Machinery, Electric Machinery, and Miscellaneous (comprising the rest of the 23 industries).

an important counterpart to the structural model in the next section, as our comparative static exercise reveals how shocks are transmitted across regions. Note that the gravity results in this section and the transmission of shocks in the structural model do not have a direct one-to-one correspondence, especially because the reduced-form regression does not focus on particular industries where the shocks arrive.

Below, we show that a straightforward OLS regression indicates that geographic distance significantly affects inter-regional demand for intermediate manufacturing. We define $DISTANCE(i, j)$ as the geographic distance between the Bureau of Economy, Trade and Industry's regional offices in regions i and j.[8] We regress $\log_{10}(T(si, hj))$—the common logarithm of the transaction value of sector s in region i sold to sector h in region j—on this distance measure, controlling for fixed effects for the sectoral pair (s, h), the home i and destination j regions, and an indicator for $i = j$ to isolate the home bias in intermediate demand for any sectoral pair. We split the sample based on whether the originating sector s is non-services (classification codes 1 to 16 in Table 6) or services (17 to 26). We also include a regression for intermediate demand from j aggregated across h, represented by M_j.

In addition, we estimate the gravity equations when the demand for si (sector s in region i) comes from consumption C_j and investment X_j in region j. In these regressions, the sample size is smaller than in the intermediate-demand regressions from hj because we lose the destination-sector dimension h.

Table 1 presents the estimation results. The benchmark estimate for $T(si, hj)$, which includes fixed effects for sectoral pairs (s, h), shows that distance has a negative effect on intermediate demand. Increasing distance by 1,000 km reduces the log transaction by 0.35 in non-service sectors, a magnitude comparable to that reported in Allen and Arkolakis (2025) for U.S. manufacturing goods. When the sample is restricted to service sectors, the estimate remains similar: -0.36. Our regression accounts for a home bias effect, $i = j$. Thus, the coefficient reflects the impact of distance on tradable goods and services only. The home bias effect is substantial, estimated at 2.01 for services and 0.84 for non-services. These estimates suggest that home demand in services far exceeds demand from outside by a factor of 10^2, while the gap between home and outside demands for non-services is less than a factor of 10. Therefore, the data imply that the flow of intermediate demand follows a geographic pattern, with neighboring regions more closely interconnected, while home bias dominates demand in the service sectors.

Estimates using the aggregated intermediate (M_j), consumption (C_j), and investment (X_j) demand also reveal an adverse impact of distance and a strong home bias for services. The negative impact of distance on M_j is more pronounced than on (hj), suggesting that the heterogeneity of sector pair (s, h) explains some of the negative effects observed in aggregate M_j. Comparing the aggregate effects of M_j, C_j, and X_j, we find that distance's influence on consumption demand is less significant than on intermediate demand. Additionally, we observe that the home bias in services is more pronounced in consumption demand than in investment demand. In the following analysis, we will use this information to interpret our numerical results on regional propagation.

[8] See https://www.meti.go.jp/english/network/regionalbureau.html for details.

Originating sector s	$\log_{10} T(si, hj)$ non	$\log_{10} T(si, hj)$ service	$\log_{10} T(si, M_j)$ non	$\log_{10} T(si, M_j)$ service	$\log_{10} T(si, C_j)$ non	$\log_{10} T(si, C_j)$ service	$\log_{10} T(si, X_j)$ non	$\log_{10} T(si, X_j)$ service
Distance (1000km)	-0.35^{***}	-0.36^{***}	-0.69^{***}	-0.53^{***}	-0.51^{***}	-0.45^{***}	-0.31^{***}	-0.21^{***}
	(0.010)	(0.011)	(0.040)	(0.052)	(0.042)	(0.058)	(0.050)	(0.068)
Home ($i = j$)	0.84^{***}	2.01^{***}	0.93^{***}	2.20^{***}	0.93^{***}	2.17^{***}	0.53^{***}	1.26^{***}
	(0.019)	(0.019)	(0.072)	(0.093)	(0.077)	(0.103)	(0.091)	(0.121)
FE i and j	✓	✓	✓	✓	✓	✓	✓	✓
FE $s \times h$	✓	✓						
FE s			✓	✓	✓	✓	✓	✓
N. obs	33696	21060	1296	810	1296	810	1296	810

Table 1: Estimates of the gravity equation. *** $p < 0.01$.

3 Model

Our model is a natural extension of a small open-economy real business cycle model to a multi-sector, multi-region setting. The economy consists of I regions. In each region, S industries operate. An industry is indexed by (s, i), where $s \in [0, S]$ and $i \in [0, I]$. In the model description, we treat s and i as real numbers, even though they are integers in the quantitative exercise.

We adopt monopolistic competition as the market structure. Because firms have pricing power, an export-demand shock translates into a terms-of-trade shock. Moreover, a similar transmission mechanism operates across regions, since each region-industry has pricing power. Monopolistic competition is common in international trade models and also allows us to analyze situations with sticky prices.

We assume each industry (s, i) is monopolistic, with only one firm producing product (s, i). Producing a good requires capital, labor, and intermediate goods. As we will detail later, we assume that capital and labor are not mobile across regions; that is, industries in region i must use the capital and labor supplied in that region. Intermediate goods are mobile; that is, a firm in region i can use intermediate inputs from any region.

As in the standard real business cycle model, each region has a single representative consumer. The representative consumer maximizes the discounted sum of instantaneous utility over an infinite horizon, making consumption, saving, and labor supply decisions. Firms make static production decisions, hiring labor, renting capital, and purchasing intermediate goods. Each firm's output is used for consumption, investment, exports, and the production of intermediate goods.

3.1 Representative consumer

The representative consumer in region i solves the utility-maximization problem

$$\sum_{t=0}^{\infty} \frac{1}{(1+\rho)^t} \left[\frac{(C_{i,t})^{1-\sigma_c} - 1}{1-\sigma_c} - \chi_i \frac{(N_{i,t})^{1+\zeta}}{1+\zeta} \right] \tag{1}$$

subject to

$$\begin{aligned} \int_0^S \int_0^I p_{sj,t} c^i_{sj,t} dj ds + \int_0^S p_{sf,t} c^i_{sf,t} ds + \int_0^S \int_0^I p_{sj,t} x^i_{sj,t} dj ds + \int_0^S p_{sf,t} x^i_{sf,t} ds \\ \leq \int_0^S w_{si,t} n_{si,t} ds + r_{i,t} K_{i,t} + \Pi_{i,t} + B_{i,t} \end{aligned} \tag{2}$$

and

$$K_{i,t+1} = (1-\delta)K_{i,t} + X_{i,t}, \tag{3}$$

where

$$C_{i,t} = \left[\int_0^S \int_0^I (\xi^i_{sjc})^{\frac{1}{\sigma}} (c^i_{sj,t})^{\frac{\sigma-1}{\sigma}} djds + \int_0^S (\xi^i_{sfc})^{\frac{1}{\sigma}} (c^i_{sf,t})^{\frac{\sigma-1}{\sigma}} ds\right]^{\frac{\sigma}{\sigma-1}},$$

$$X_{i,t} = \left[\int_0^S \int_0^I (\xi^i_{sjx})^{\frac{1}{\sigma}} (x^i_{sj,t})^{\frac{\sigma-1}{\sigma}} djds + \int_0^S (\xi^i_{sfx})^{\frac{1}{\sigma}} (x^i_{sf,t})^{\frac{\sigma-1}{\sigma}} ds\right]^{\frac{\sigma}{\sigma-1}},$$

and

$$N_{i,t} = \left[\int_0^S (n_{si,t})^{\frac{\tau+1}{\tau}} ds\right]^{\frac{\tau}{\tau+1}}.$$

Here, $C_{i,t}$ is the consumption of composite goods in period t. The price of good (s,j) in period t is $p_{sj,t}$, and it is assumed to be common across destination regions i. The prices of the imported goods are assumed to be exogenously given at $p_{sf,t}$. The notation c^i_{sj} represents the consumption of good s from region j by agent i, while ξ^i_{sjc} is a parameter. The consumption goods with c^i_{sf} are imported goods, where f represents "foreign." The imported goods have S varieties, which are common across regions and exogenously given. The variable N_i is the composite labor supply. The consumer supplies labor n_{si} for the production of good s. The labor market is perfectly competitive, and the wage rate is w_{si} for industry (s,i). The variable $\Pi_{i,t}$ is the profit of the firms in region i. $B_{i,t}$ denotes an international transfer exogenous to the households in i. The variable $X_{i,t}$ is the investment in composite goods by consumer i in period t. The notation x^i_{sj} represents the investment in good s from region j by region i in period t, and ξ^i_{sjx} is a time-invariant parameter. $K_{i,t}$ is the capital stock in region i and period t, which is augmented via the region-specific investment good $X_{i,t}$ and depreciates at rate δ every period.

To consume the goods, the consumer allocates consumption across goods by solving the expenditure-minimization problem each period t:

$$\min_{\{c^i_{sj,t}\}_{sj},\{c^i_{sf,t}\}_s} \int_0^S \int_0^I p_{sj,t} c^i_{sj,t} djds + \int_0^S p_{sf,t} c^i_{sf,t} ds$$

subject to

$$C_{i,t} = \left[\int_0^S \int_0^I (\xi^i_{sjc})^{\frac{1}{\sigma}} (c^i_{sj,t})^{\frac{\sigma-1}{\sigma}} djds + \int_0^S (\xi^i_{sfc})^{\frac{1}{\sigma}} (c^i_{sf,t})^{\frac{\sigma-1}{\sigma}} ds\right]^{\frac{\sigma}{\sigma-1}}.$$

The solution of the optimization implies the demand for domestic goods

$$c^i_{sj,t} = \left(\frac{p_{sj,t}}{P^c_{i,t}}\right)^{-\sigma} \xi^i_{sjc} C_{i,t},$$

and for imported foreign goods

$$c^i_{sf,t} = \left(\frac{p_{sf,t}}{P^c_{i,t}}\right)^{-\sigma} \xi^i_{sfc} C_{i,t}, \tag{4}$$

where the price index $P^c_{i,t}$ is written as

$$P^c_{i,t} := \left[\int_0^S \int_0^I \xi^i_{sjc} (p_{sj,t})^{1-\sigma} djds + \int_0^S \xi^i_{sfc} (p_{sf,t})^{1-\sigma} ds\right]^{\frac{1}{1-\sigma}}. \tag{5}$$

Similarly, the household's minimization of investment costs yields

$$x^i_{sj,t} = \left(\frac{p_{sj,t}}{P^x_{i,t}}\right)^{-\sigma} \xi^i_{sjx} X_{i,t}$$

for $j \in \{[0,I], f\}$, where the price index $P^x_{i,t}$ is given as

$$P^x_{i,t} := \left[\int_0^S \int_0^I \xi^i_{sjx} (p_{sj,t})^{1-\sigma} dj ds + \int_0^S \xi^i_{sfx} (p_{sf,t})^{1-\sigma} ds\right]^{\frac{1}{1-\sigma}}.$$

The consumer's budget constraint can now be rewritten as

$$P^c_{i,t} C_{i,t} + P^x_{i,t} X_{i,t} \leq \int_0^S w_{si,t} n_{si,t} ds + r_{i,t} K_{i,t} + \Pi_{i,t} + B_{i,t}.$$

The consumer's intertemporal optimization implies the Euler equation and the labor supply relationship

$$\left(\frac{C_{i,t}}{C_{i,t+1}}\right)^{-\sigma_c} = \frac{1}{1+\rho} \frac{P^c_{i,t}}{P^c_{i,t+1}} \frac{P^x_{i,t+1}}{P^x_{i,t}} \left(1 + \frac{r_{i,t+1}}{P^x_{i,t+1}} - \delta\right) \tag{6}$$

and

$$\frac{w_{si,t}}{P^c_{i,t}} = \chi_i (C_{i,t})^{\sigma_c} (N_{i,t})^{\zeta} \left(\frac{n_{si,t}}{N_{i,t}}\right)^{\frac{1}{\tau}}. \tag{7}$$

3.2 Production

In region i, good h is produced by the production function

$$y_{hi,t} = A_{hi} (M_{hi,t})^{\alpha_h} (N_{hi,t})^{\beta_h} (K_{hi,t})^{1-\alpha_h-\beta_h}, \tag{8}$$

where

$$M_{hi,t} = \left[\int_0^S \int_0^I (\gamma^{hi}_{sj})^{\frac{1}{\sigma}} (m^{hi}_{sj,t})^{\frac{\sigma-1}{\sigma}} dj ds + \int_0^S (\gamma^{hi}_{sf})^{\frac{1}{\sigma}} (m^{hi}_{sf,t})^{\frac{\sigma-1}{\sigma}} ds\right]^{\frac{\sigma}{\sigma-1}}.$$

Here, $m^{hi}_{sj,t}$ is intermediate good s from region j (and $m^{hi}_{sf,t}$ imported good s) used in production of good h in region i and period t, and γ^{hi}_{sj} is a parameter.

The demand function for intermediate goods is

$$m^{hi}_{sj,t} = \left(\frac{p_{sj,t}}{P^m_{hi,t}}\right)^{-\sigma} \gamma^{hi}_{sj} M_{hi,t},$$

for $j \in \{[0,I], f\}$, where

$$P^m_{hi,t} := \left[\int_0^S \int_0^I \gamma^{hi}_{sj} (p_{sj,t})^{1-\sigma} dj ds + \int_0^S \gamma^{hi}_{sf} (p_{sf,t})^{1-\sigma} ds\right]^{\frac{1}{1-\sigma}}. \tag{9}$$

Thus, the total demand for good (s,j) in period t is, by adding the consumption demand, investment demand, and intermediate good demand,

$$y_{sj,t} = \int_0^I (c^i_{sj,t} + x^i_{sj,t}) di + \int_0^S \int_0^I m^{hi}_{sj,t} di dh + y^f_{sj,t} \tag{10}$$

$$= (p_{sj,t})^{-\sigma} \left(\int_0^I \left((P^c_{i,t})^{\sigma} \xi^i_{sjc} C_{i,t} + (P^x_{i,t})^{\sigma} \xi^i_{sjx} X_{i,t}\right) di + \int_0^S \int_0^I (P^m_{hi,t})^{\sigma} \gamma^{hi}_{sj} M_{hi,t} di dh\right) + y^f_{sj,t}, \tag{11}$$

where $y^f_{sj,t}$ represents the foreign (export) demand in period t. Assume that foreign demand takes the form

$$y^f_{sj,t} = \omega^f_{sj,t}(p_{sj,t})^{-\sigma}(\bar{P}_t)^{\sigma}, \tag{12}$$

that is, foreign demand has the same price elasticity as domestic demand. $\bar{P}$ is the price level in the foreign country.

The demand for good (s, j) can be expressed as $(p_{sj,t})^{-\sigma}D_{sj,t}$ where $D_{sj,t}$ is a demand shifter:

$$D_{sj,t} := \left(\int_0^I \left((P^c_{i,t})^{\sigma}\xi^i_{sjc}C_{i,t} + (P^x_{i,t})^{\sigma}\xi^i_{sjx}X_{i,t}\right)di + \int_0^S\int_0^I (P^m_{hi,t})^{\sigma}\gamma^{hi}_{sj}M_{hi,t}didh\right) + \omega^f_{sj,t}(\bar{P}_t)^{\sigma}. \tag{13}$$

We analyze the firm's problem in two steps. First, the firm chooses the combination of inputs to minimize the unit cost:

$$\min_{M_{sj,t},N_{sj,t},K_{sj,t}} \; P^m_{sj,t}M_{sj,t} + w_{sj,t}N_{sj,t} + r_{j,t}K_{sj,t}$$

subject to

$$1 = A_{sj}(M_{sj,t})^{\alpha_s}(N_{sj,t})^{\beta_s}(K_{sj,t})^{1-\alpha_s-\beta_s}.$$

The solution yields the unit cost $\lambda_{sj,t}$:

$$\lambda_{sj,t} = \frac{(P^m_{sj,t})^{\alpha_s}(w_{sj,t})^{\beta_s}(r_{j,t})^{1-\alpha_s-\beta_s}}{A_{sj}\alpha_s^{\alpha_s}\beta_s^{\beta_s}(1-\alpha_s-\beta_s)^{1-\alpha_s-\beta_s}} \tag{14}$$

and the derived factor demand for unit output:

$$M_{sj,t} = \frac{\alpha_s}{P^m_{sj,t}}\lambda_{sj,t},$$

$$N_{sj,t} = \frac{\beta_s}{w_{sj,t}}\lambda_{sj,t},$$

and

$$K_{sj,t} = \frac{1-\alpha_s-\beta_s}{r_{j,t}}\lambda_{sj,t}.$$

Second, the firm maximizes profit:

$$\max_{p_{sj,t}} \; (p_{sj,t} - \lambda_{sj,t})(p_{sj,t})^{-\sigma}D_{sj,t}. \tag{15}$$

The result is the standard constant markup rule:

$$p_{sj,t} = \frac{\sigma}{\sigma-1}\lambda_{sj,t}. \tag{16}$$

Thus the production of good (s, j) is

$$y_{sj,t} = \left(\frac{\sigma}{\sigma-1}\lambda_{sj,t}\right)^{-\sigma} D_{sj,t}. \tag{17}$$

The derived factor demand can, therefore, be computed from:

$$M_{sj,t} = \frac{\alpha_s}{P^m_{sj,t}}\lambda_{sj,t}y_{sj,t}, \tag{18}$$

$$N_{sj,t} = \frac{\beta_s}{w_{sj,t}}\lambda_{sj,t}y_{sj,t}, \tag{19}$$

and

$$K_{sj,t} = \frac{1-\alpha_s-\beta_s}{r_{j,t}}\lambda_{sj,t}y_{sj,t}. \tag{20}$$

3.3 Trade balance

We allow trade imbalances at both the national and regional levels. In our model, trade imbalances are possible only when international transfers exist. We denote by $B_{i,t}$ an (exogenous) international transfer to households in region i at time t. The international trade account in our model is written as

$$\int_0^I B_{i,t}di = \int_0^S \int_0^I p_{sf,t}\left(c^i_{sf,t} + x^i_{sf,t} + \int_0^S m^{hi}_{sf,t}dh\right)dids - \int_0^S \int_0^I p_{si,t}y^f_{si,t}dids. \tag{21}$$

The first term is the value of imports, where $p_{sf,t}$ is the price of import good s, which is exogenous under the small open economy assumption, and $(c^i_{sf,t}, x^i_{sf,t}, m^{hi}_{sf,t})$ are the consumption, investment, and industry h's intermediate demand by region i for import good s. The second term on the right-hand side is the value of exports, where $p_{si,t}$ is the price of good s from region i and $y^f_{si,t}$ is the export demand for good s from region i.[9]

3.4 Equilibrium

We treat the foreign-good prices $p_{sf,t}$ for all s, t as exogenous. Given the prices of domestic and foreign goods, factor prices, and profit income, consumers in region i at time t demand consumption goods $(c^i_{sj,t}, c^i_{sf,t})$ for all j and investment goods $(x^i_{sj,t}, x^i_{sf,t})$ for all j by optimizing (1).

Firms in region j at time t face demand from consumers and foreign countries as a function of $p_{sj,t}$. Given factor prices, each firm sets the price of its good to maximize profit (15).

Markets clear for each good in each region. The total demand for good s in region i is given by (11), with export demand given by (12). The total supply for good h in region i is given by (8). The market-clearing conditions for labor and capital require:

$$N_{si,t} = n_{si,t},$$

and

$$\int_0^S K_{si,t}ds = K_{i,t}$$

for all (s, i) and i. The profit income in equilibrium must satisfy

$$\Pi_{i,t} = \int_0^S (p_{si,t} - \lambda_{si,t})y_{si,t}ds. \tag{22}$$

Note that once the goods and factor markets clear, the trade balance condition (21) is automatically satisfied. This follows from aggregating the consumer's budget constraint (2) and imposing the market-clearing conditions. Later in the quantitative analysis, we will set $B_{i,t}$ to a value that fits the data.

4 Basic mechanism: an analysis of a symmetric static model

In this section, we analyze a special case of the model to build intuition about how a foreign demand shock affects the aggregate economy. We make three main points. First, a foreign demand shock

[9] We abstract from net primary income, as it remained relatively stable over the period of interest. For example, net primary income amounted to 3.6% of GDP in 2008Q3 and remained at 3.2% in 2009Q1 (relative to 2008Q3 GDP), when GDP fell sharply. This stands in contrast to net exports, which declined from 4.3% of GDP in 2008Q3 to −1.2% in 2009Q1.

increases output and domestic prices relative to foreign prices, thereby generating a terms-of-trade shock. Second, there is a multiplier effect. Third, price changes dampen the effects of shocks.

For analytical tractability, we abstract from saving and capital accumulation and analyze a static model. The full static model is described in Appendix F. In this section, we further assume symmetry across sectors and regions, that is, goods from all sector-regions enter the CES aggregators symmetrically ($\xi^i_{sj} = \gamma^{hi}_{sj} = 1, \forall h, i, j, s$), the import weights for final and intermediate goods are symmetric ($\xi^i_{sf} = \xi_f$ and $\gamma^{hi}_{sf} = \gamma_f$, $\forall h, i, s$), intermediate cost shares are constant across industries ($\alpha_s = \alpha, \forall s$), and that international trade balances. In addition, to simplify the expressions, we normalize $S = I = 1$ and assume symmetric productivity across regions: $A^{hi} = (\alpha^\alpha (1-\alpha)^{1-\alpha})^{-1}, \forall h, i$.

In this symmetric environment, the consumer's demand for domestic and foreign goods can be expressed as

$$c^i_{sj} = c = \left(\frac{p}{P}\right)^{-\sigma} C$$

and

$$c^i_{sf} = c_f = \left(\frac{p_f}{P}\right)^{-\sigma} \xi_f C,$$

where the price index is

$$P = (p^{1-\sigma} + \xi_f p_f^{1-\sigma})^{1/(1-\sigma)} \tag{23}$$

and aggregate consumption is

$$C = (c^{(\sigma-1)/\sigma} + \xi_f^{1/\sigma} c_f^{(\sigma-1)/\sigma})^{\sigma/(\sigma-1)}.$$

The intermediate-good demand for domestic goods is

$$m = \left(\frac{p}{P_m}\right)^{-\sigma} M$$

and that for foreign goods is

$$m_f = \left(\frac{p_f}{P_m}\right)^{-\sigma} \gamma_f M,$$

where

$$P_m = (p^{1-\sigma} + \gamma_f p_f^{1-\sigma})^{1/(1-\sigma)}. \tag{24}$$

Thus, $M = (m^{(\sigma-1)/\sigma} + \gamma_f^{1/\sigma} m_f^{(\sigma-1)/\sigma})^{\sigma/(\sigma-1)}$. We assume foreign demand is $y^f = p^{-\sigma}\omega^f$, where ω^f is the foreign demand parameter. The total demand for a good is

$$y = p^{-\sigma}(P^\sigma C + P_m^\sigma M + \omega^f).$$

The monopolist's optimal price follows the markup rule:

$$p = \frac{\sigma}{\sigma - 1}\lambda,$$

where λ is the unit cost

$$\lambda = P_m^\alpha w^{1-\alpha}.$$

Because the final-good sector's production function is $y = AM^\alpha N^{1-\alpha}$, the factor demands of final-good firms are

$$M = \frac{\alpha \lambda y}{P_m}$$

and

$$N = \frac{(1-\alpha)\lambda y}{w}.$$

Hence, $M/N = (\alpha/(1-\alpha))(w/P_m)$. The household budget constraint implies $PC = py - P_m M = \mu_c py$, where $\mu_c := 1 - \alpha(\sigma-1)/\sigma$ is the share of consumption expenditure in output sales. Finally, the labor supply function, derived from the household's optimization, is

$$\frac{w}{P} = \chi C^{\sigma_c} N^{\zeta}.$$

Let TS denote the export share of total output sales, and let TS_c and TS_m denote the import shares of consumption and intermediate expenditure, respectively. We obtain the following result.

Proposition 1 *(a) Suppose $\sigma_c \leq 1$ and $\gamma_f > 0$. An equilibrium exists and is unique for each $\omega^f > 0$, and the equilibrium p/P decreases continuously to 1 as ω^f decreases to 0. Equilibrium consumption C, output y, real wages w/P, and the price p of domestically produced goods relative to imported goods are strictly increasing in ω^f.*

(b) Suppose $\sigma_c = 1$. The export-demand shock multiplier on value added is given by

$$\frac{d(\mu_c py)}{d(p^{1-\sigma}\omega^f)} = \frac{\mu_c}{TS + \left(\mu_c \frac{1-TS_c}{1-TS_m}\frac{TS_c}{TS_m} + 1 - \mu_c\right)\frac{(\sigma-1)(1-TS_m)}{\frac{1}{TS_m}+\frac{\alpha}{1-\alpha}}} > 0 \tag{25}$$

Proof: See Appendix D.

With definitions of $P(p)$ and $P_m(p)$ in (23) and (24), $\Gamma(p) := p/P(p)$ and $\Gamma_m(p) := p/P_m(p)$ are increasing in p. The export share of output sales, $TS(p) = 1 - \left(\mu_c \Gamma(p)^{1-\sigma} + (1-\mu_c)\Gamma_m(p)^{1-\sigma}\right)$, is also increasing in p.

In Appendix D, we show that the equilibrium (p, y) is determined by the following two equations:

$$y = \frac{p^{-\sigma}\omega^f}{TS(p)}, \tag{26}$$

$$y^{\sigma_c+\zeta} = y_o \left(\frac{p}{P}\right)^{1-\sigma_c} \left(\frac{p}{P_m}\right)^{\frac{\alpha(1+\zeta)}{1-\alpha}}, \tag{27}$$

where $y_o := ((\sigma-1)/\sigma)^{\frac{1+\alpha\zeta}{1-\alpha}} / (\chi(\mu_c)^{\sigma_c}(1-\alpha)^{\zeta})$ is a constant.

Figure 7 plots (26) as a downward-sloping curve in the (y, p) plane. The demand for domestic goods declines as the domestic price p rises because export demand in the numerator falls and the export share of sales, $1 - (\mu_c(p/P)^{1-\sigma} + (1-\mu_c)(p/P_m)^{1-\sigma})$, in the denominator rises. Thus, the first equation is interpreted as a demand curve for domestic goods as a function of the terms of trade p/p_f.

Equation (27) represents a supply curve, i.e., the response of domestic production to p. It is upward-sloping if $\sigma_c \leq 1$ and $\gamma_f > 0$, and it may be upward-sloping if $\sigma_c > 1$ when the share of intermediate imports γ_f is sufficiently large.

Only the demand curve shifts with ω^f. Figure 7 shows the positive effects of an export shock ω^f on the domestic price p and output y in the equilibrium established by Proposition 1(a). The equilibrium response of y depends on the slope of the supply curve.

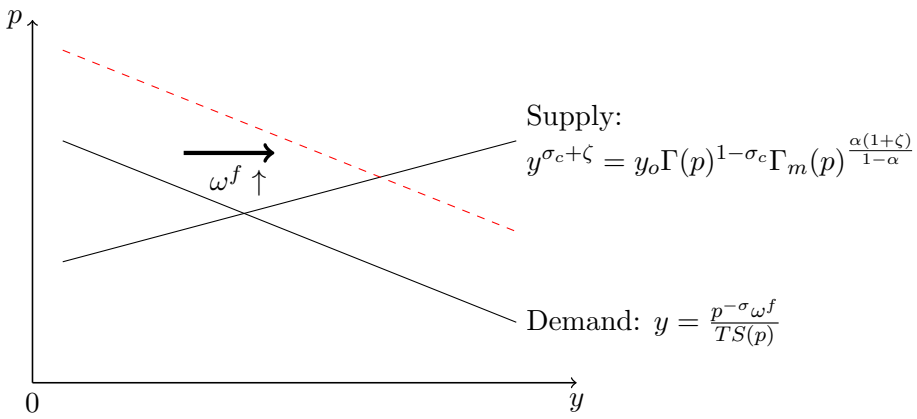


Figure 7: Demand and supply for domestically produced goods. $\Gamma(p)$, $\Gamma_m(p)$, and $TS(p)$ are increasing functions that satisfy $\Gamma(p) = p/P$, $\Gamma_m(p) = p/P_m$, and $TS(p) = 1 - \left(\mu_c\Gamma(p)^{1-\sigma} + (1-\mu_c)\Gamma_m(p)^{1-\sigma}\right)$ in equilibrium.

In the special case when $\sigma_c = 1$ and $\gamma_f = 0$, the supply curve is vertical because equilibrium employment does not change, as the income and substitution effects cancel out, and because no propagation channel exists through imported intermediate goods. So ω^f has no effect on y. Even in this case, p increases, and hence py and C strictly increase with ω^f. The import share of consumption increases because the relative import price p_f/p decreases.

Here, the price response through monopolistic competition is essential in the mechanism. An increase in foreign demand raises the monopoly price of domestically produced goods. This price increase improves the terms of trade p/p_f, which in turn affects the rest of the economy, including real wages. The mechanism is reminiscent of the demand externality in a monopolistically competitive economy with international trade, as in Matsuyama (1992), although our model lacks increasing returns to scale.

Proposition 1(b) derives the multiplier effect of export demand on GDP for the case $\sigma_c = 1$. When $\sigma_c = 1$, labor supply is constant, and output supply y depends on intermediates only through $\Gamma_m(p)^{\alpha/(1-\alpha)}$. Equilibrium p is determined by

$$p \cdot y(p) \cdot TS(p) = p^{1-\sigma}\omega^f.$$

Thus, we obtain the multiplier effect of $p^{1-\sigma}\omega^f$ on $\mu_c py$ in (25) as the inverse of the export share, attenuated by the decline in the domestic demand share. The multiplier (25) simplifies in the special case $\xi_f = \gamma_f$, where $P = P_m$, $TS_c = TS_m = TS$, and thus:

$$\frac{d(\mu_c py)}{d(p^{1-\sigma}\omega^f)} = \frac{(1-\alpha+\alpha/\sigma)/TS}{1 + \frac{(\sigma-1)(1-TS)}{1+\frac{\alpha}{1-\alpha}TS}}. \tag{28}$$

This expression shows the main multiplier effect of the export share of GDP in the numerator and the attenuation effect determined by (σ, α, TS) in the denominator.

The multiplier (28) converges to zero as $\sigma \to \infty$. In a competitive economy, the demand shock does not affect income because the equilibrium price system remains unchanged. This again shows that monopolistic competition is essential for generating an amplification effect from the demand shock.

The multiplier can be substantial. For a back-of-the-envelope calibration of the Japanese economy in 2008, we set $\sigma = 5$ and the export share of GDP to 17%. We also set the share of intermediate goods in the static model to $\alpha = 0.5$. This calibration yields a multiplier of 1.38 for (28). This value is comparable to what Japan experienced in 2008, under the assumption that the export shock caused the GDP decline: an 8.8% decline in GDP for a 36.1% decline in exports yields a multiplier of $8.8/(36.1 \times 0.17) = 1.43$. According to our formula, the model multiplier falls below one to 0.79 when the export share rises to 35%. This result suggests that the large multiplier effect comes from a small export share.

Another example is the case with no intermediate imports ($\gamma_f = 0$), in which the export share reduces to $\mu_c(1 - (p/P)^{1-\sigma})$. This value is smaller than in the case $\gamma_f = \xi_f > 0$, where the export share is $1 - (p/P)^{1-\sigma}$. This result indicates a leakage of the multiplier effect when the import share is large. Japan's relatively small exposure to trade and its pricing power over its export products generate a large multiplier for export shocks, leading to the familiar pattern of "when the U.S. sneezes, Japan catches a cold."

In the next section, we return to the original dynamic model and quantitatively assess the effect of the export shock during the Great Recession. We will see that the basic mechanisms we described here also operate in the more complex quantitative model.

5 Computation and calibration

We quantify the model for the Japanese economy. First, we provide a brief outline of the computational method. Then, we describe how we calibrate the baseline economy.

5.1 Computation

Here, we outline the computation of our quantitative model. The details of the computational procedure are provided in Appendix G.

First, we compute the model's steady state. We assume the initial steady state is the Japanese economy in 2008Q3 and compute the model's steady state with constant parameter values so that the model fits the data in 2008Q3. This steady state serves as the initial point of our experiment.

We assume that the export shock arrives as an unanticipated change in $\omega^f_{sj,t}$ for $t = 1$ to $t = T$, where $t = 1$ is 2008Q4 and $t = T$ is 2010Q4. That is, until 2008Q3 ($t = 0$), economic agents believe the economy will remain in the initial steady state forever. At the beginning of $t = 1$, new information about future $\omega^f_{sj,t}$ ($t = 1, 2, \ldots$) arrives. As we will see in the next section, we set $\omega^f_{sj,t}$ from $t = 1$ to $t = T$ so that the realized export values during that period match the corresponding data from Japan for all (s, j, t). We assume that the economy reaches its final steady state in 2010Q4 ($t = T$), with $\omega^f_{sj,t}$ remaining constant thereafter.

We assume that, at time $t = 1$, all future time series for $\omega^f_{sj,t}$ from $t = 2008Q4$ onward are revealed to economic agents. Therefore, after $t = 1$, the economy follows perfect-foresight transition dynamics.

5.2 Calibration

We assume that one period is equivalent to one quarter. The baseline economy is in a steady state, with the 2008Q3 outcome. We set the baseline parameter values to ensure that the equilibrium

outcome aligns with statistics for the Japanese economy at that time.

We follow a standard two-step calibration strategy. In the first step, we set a subset of parameters—such as factor shares and the capital depreciation rate—using values commonly used in the literature or directly observed in the data. In the second step, we calibrate the remaining parameters endogenously within the model to match targeted moments computed from the data. In total, the model requires 65,856 parameters, of which 65,772 are determined endogenously. Below, we describe how each parameter is determined.

The consumption share parameters $\{\xi^i_{sjc}\}_{i,sj}$ are calibrated so that the consumption expenditure share of sj (good s produced in region j) by the region i consumer matches the data in the baseline economy. The consumption shares are taken from the inter-regional input-output table in 2005 (IRIO2005), the closest time period before 2008Q3. Parameters governing the demand for sj production by foreign countries, $\{\omega^f_{sj}\}_{sj}$, are set so that the GDP share of sj exports matches the data computed in IRIO2005.[10]

In the baseline economy, we assume that the parameter governing the wage elasticity of labor supply in each industry, τ, equals 1, as in Horvath (2000). The inverse of the Frisch elasticity of overall labor supply, ζ, is set to 2.5, based on the empirical estimate by Kuroda and Yamamoto (2008).[11] The labor disutility parameter χ_i is calibrated to replicate the regional variation in the employed population in 2008Q3, drawn from the LFS conducted by the Statistics Bureau of Japan.[12] The employed population reflects the labor force and the employment rate, while variation in working hours per labor force is not considered here because reliable regional data on working hours are unavailable. The time discount rate ρ is set to 0.01. As a benchmark, we consider the log utility case, $\sigma_c \to 1$.

The investment share parameters $\{\xi^i_{sjx}\}_{i,sj}$ are calibrated so that the investment expenditure share of each sj in region i in the benchmark economy matches the IRIO2005 data. The elasticity-of-substitution parameter, σ, is set to 5.0 in the baseline economy. The parameters governing the cost share of each intermediate good sj for the producer of good h in region i, $\{\gamma^{hi}_{sj}\}_{hi,sj}$, are set so that the benchmark values match the IRIO2005 counterparts. The factor-neutral productivity for each industry sj, A_{sj}, is given by the product of the industry- and region-specific productivity parameters; that is, $A_{sj} = A_s \times A_j$, where A_s is the industry-specific productivity and A_j is the region-specific productivity. First, we map the industry classification in the JIP Database to ours and compute the industrial TFP and the cost share of intermediate goods for each industry s ($\{\alpha_s\}_s$). Given all other parameters, the region-specific productivity A_j is pinned down so that the regional variation in the average wage rate in the benchmark replicates the data counterpart computed using the MLS.

We set the international transfer levels to each region in the initial economy, $\{B_{i,0}\}_{i\in I}$, to match the national net exports in 2008Q3, which totaled 4.32% of national GDP. We also set the weight parameters for imported goods, $\{\xi^i_{sfc}\}_{i,s}$, $\{\xi^i_{sfx}\}_{i,s}$, and $\{\gamma^{hi}_{sf}\}_{hi,s}$, using strategies similar to those used to pin down the weight parameters for domestically produced goods. Further details on the calibration procedure are provided in Appendix C.

Table 2 summarizes the parameter values. The values of A_s and $\{\alpha_s, \beta_s\}_s$ are presented in

[10] As a result, the export-to-GDP ratio and the share of sj in total exports match the data.

[11] Kuroda and Yamamoto (2008) estimate the Frisch elasticity in Japan and report that the elasticity on the extensive and intensive margins combined ranges between 0.2 and 0.7 for males. $\zeta = 2.5$ implies a Frisch elasticity of 0.4.

[12] See https://www.stat.go.jp/english/index.html.

Parameter	Description	Value	Target/Source
Preference			
ρ	Time discount rate	0.01	Assumed
σ_c	Curvature for utility	1.0	Assumed
ζ	Inverse of Frisch elasticity	2.5	Kuroda and Yamamoto (2008)
τ	Elasticity of substitution (labor)	1.0	Benchmark in Horvath (2000)
$\{\chi_i\}_i$	Disutility of labor supply	Table 7	Regional labor supply/LFS (2008)
$\{\xi^i_{sjc}\}_{i,sj}, \{\xi^i_{sfc}\}_{i,s}$	Weight on consumption goods	Fig. 19a	Consumption share/IRIO (2005)
$\{\omega^f_{sj}\}_{sj}$	Weight on export goods	Fig. 19d	Export share/IRIO (2005)
Technology			
σ	Elasticity of substitution	5.0	Assumed
δ	Capital depreciation rate	0.015	Assumed
$\{\alpha_s\}_s$	Cost share of intermediate goods	Table 6	Taken from JIP (2008)
$\{\beta_s\}_s$	Labor share	Table 6	Taken from JIP (2008)
$\{A_s \times A_j\}_{sj}$	Factor-neutral productivity		
$\{A_s\}_s$	Industry-specific	Table 6	Taken from JIP (2008)
$\{A_j\}_j$	Region-specific	Table 7	Regional wage variation/MLS (2008)
$\{\gamma^{hi}_{sj}\}_{hi,sj}, \{\gamma^{hi}_{sf}\}_{hi,s}$	Weight on intermediate goods	Fig. 19c	Intermediate share/IRIO (2005)
$\{\xi^i_{sjx}\}_{i,sj}, \{\xi^i_{sfx}\}_{i,s}$	Weight on investment goods	Fig. 19b	Investment share/IRIO (2005)

Table 2: Summary of the parameter values, their source/reference, and data for setting targets.

Table 6 in Appendix C. The regional parameters A_i and χ_i are presented in Table 7 in Appendix C. The parameters $\{\xi^i_{sjc}\}_{i,sj}$, $\{\xi^i_{sjx}\}_{i,sj}$, $\{\gamma^{hi}_{sj}\}_{hi,sj}$, and $\{\omega^f_{sj}\}_{sj}$ are too numerous to summarize in a table and are shown as heatmaps in Figure 19 in Appendix C.

6 Simulating the model with the Great Recession export shocks

The primary purpose of our quantitative model is to analyze how export shocks propagate during the Great Recession. As mentioned earlier, we set the export demand parameter $\omega^f_{sj,t}$ so that the resulting export time series matches the data. More precisely, we simulate export shocks to industry si (i.e., industry s in region i) in period t by adjusting $y^f_{si,t}$ so that the following holds in equilibrium:

$$\frac{y^f_{si,t}}{y^f_{si,t=0}} = \frac{\text{real export of } si \text{ in } t \text{ in data}}{\text{real export of } si \text{ in } t=0 \text{ in data}}, \tag{29}$$

where the reference period $t = 0$ corresponds to 2008Q3.

A variable of primary interest is domestic final demand, composed of consumption and investment. The *real* consumption is computed excluding consumption of imported goods. Formally, the real consumption for region i in period t, $\bar{C}_{i,t}$, is defined as

$$\bar{C}_{i,t} = \int_0^S \int_0^I p_{sj,t=0} c^i_{sj,t} dj ds, \tag{30}$$

where $p_{sj,t=0}$ is the price of goods produced in industry sj in the reference period $t = 0$. Similarly,

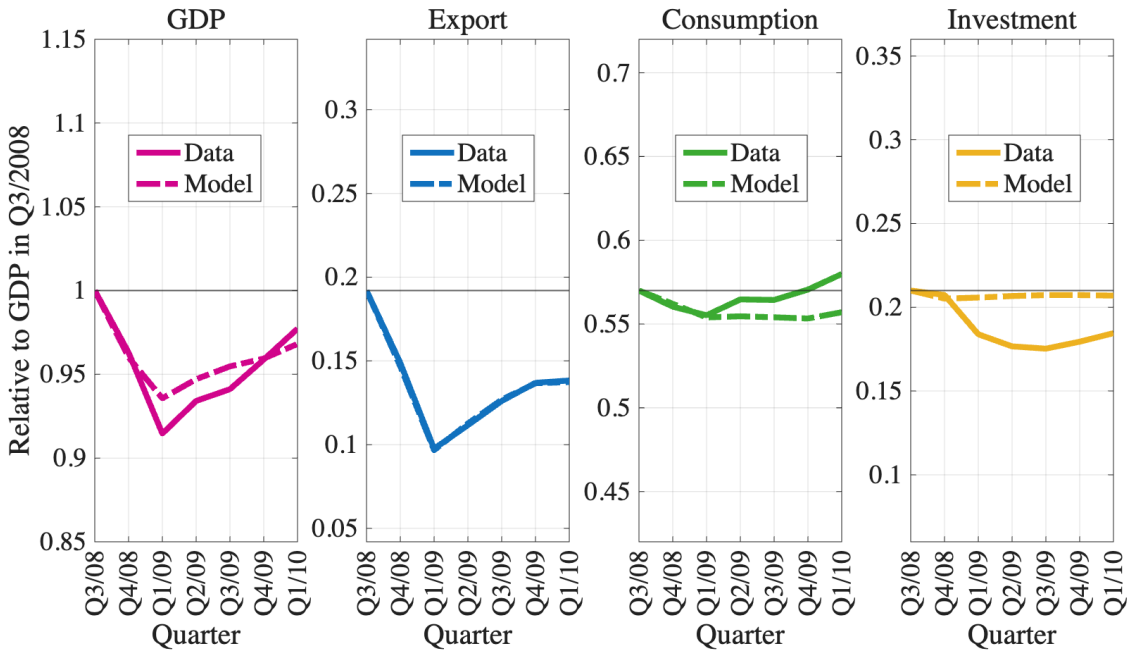


Figure 8: National responses. Each series is expressed relative to GDP in 2008Q3. "Data" plots fluctuations in HP-filtered variables. Consumption data include only private consumption and exclude government consumption. A similar construction applies to investment.

the real investment for region i in period t, $\bar{X}_{i,t}$, is defined as

$$\bar{X}_{i,t} = \int_0^S \int_0^I p_{sj,t=0} x^i_{sj,t} djds. \tag{31}$$

The national real consumption and investment in period t, $\bar{C}_{Japan,t}$ and $\bar{X}_{Japan,t}$, are defined as

$$\bar{C}_{Japan,t} = \int_0^I \bar{C}_{i,t} di \tag{32}$$

and

$$\bar{X}_{Japan,t} = \int_0^I \bar{X}_{i,t} di. \tag{33}$$

6.1 National level response

Figure 8 shows the time series of GDP, exports, consumption, and investment at the national level, comparing the model and the data. Consumption data include only private consumption and exclude government consumption. The same construction applies to investment. By construction, the model's export values match the data exactly.

The consumption and investment time series are not our targets, so our procedure does not guarantee that they will fit the data. The model implies a slightly larger decline in consumption than is observed in the data, and the recovery is also slower in the model. One potential explanation for the discrepancy is that we do not capture the increased government spending observed in the data during those periods.[13] Increased government spending would have raised household disposable

[13]For example, the sum of government consumption and investment increased by 12% in the first quarter of 2009 compared with the third quarter of 2008, corresponding to 3.3% of the domestic demand in the first quarter of 2008.

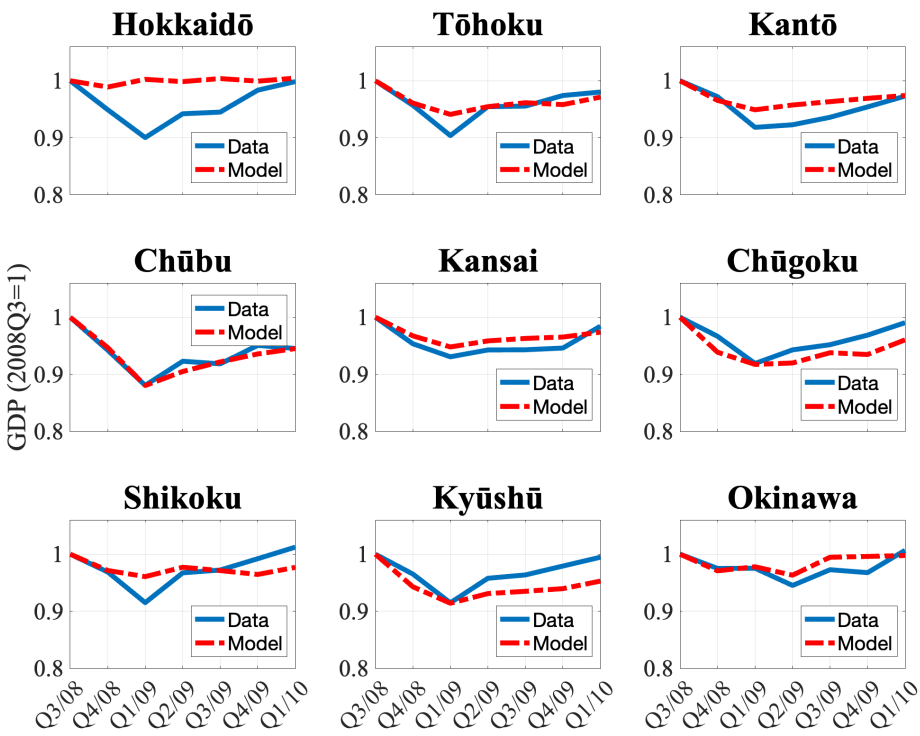


Figure 9: Regional responses

income and mitigated the consumption decline to some extent. Thus, abstracting from government spending in the model would lead to a larger decline in (private) consumption than is observed in the data.

The model's investment response is significantly smaller. This outcome suggests that investment movements during this period were primarily driven by factors outside the scope of this model. Another possible reason is that our assumption of perfect foresight attenuates the investment response. That is, because economic agents know that the recovery of exports is relatively quick, investment (which would respond more to a permanent shock) does not adjust much in the model. In reality, the duration of the export shock was uncertain, and firms may have perceived it as potentially more persistent.

The overall response of output is comparable to the data. The model response was smaller, reflecting the large discrepancy in the investment response. The persistent decline in consumption and the smaller investment response offset each other, bringing the model's GDP recovery close to the data. In the next section, we examine the propagation of the export shock across regions in greater detail.

6.2 Regional level responses

Figure 9 compares the model's regional demand for domestic final goods with the data. The model fits the data particularly well for regions such as Chūbu, Chūgoku, and Kyūshū, which experienced large declines in exports. In contrast, for regions such as Hokkaidō, Tōhoku, and Shikoku, where exports are either a small share of regional GDP or export declines were not significant, the model

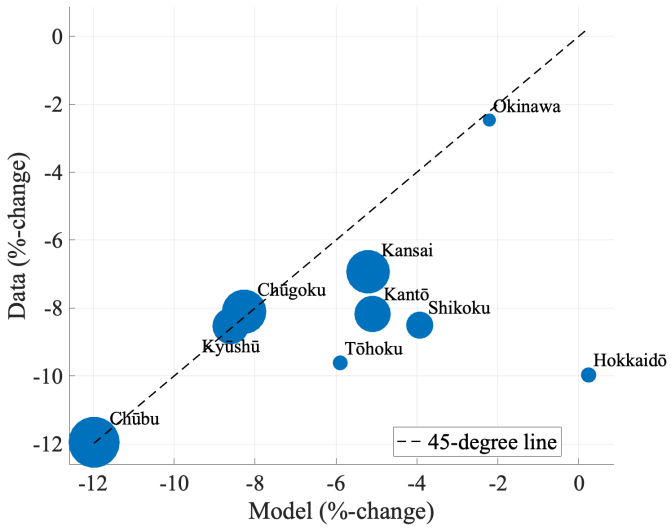


Figure 10: Regional GDP declines in model predictions and in the data. The size of the circle shows the export share of regional GDP.

poorly explains the regional GDP decline.

The model's varying explanatory power across regions is illustrated in Figure 10, which plots the observed decline in regional GDP from 2008Q3 to 2009Q1 (vertical axis) against the model's predictions (horizontal axis). The size of the markers indicates each region's export share of GDP. The scatter plots lie below the 45-degree line, indicating that the model generally underestimates the decline in GDP. The model nearly accounts for all GDP declines in regions with high export shares, such as Chūbu, Kyūshū, and Chūgoku. In contrast, the model explains a smaller share of the decline in regions with low export shares. However, the presence of regions such as Tōhoku, where the model's predicted decline is substantial despite the small share of exports, indicates inter-regional propagation of shocks.

Among the regions with small model declines, Shikoku, Hokkaidō, and Tōhoku account for small shares of national GDP (2.7%, 3.8%, and 6.1%, respectively, in the 2005 IRIO). Their movements are relatively inconsequential for aggregate GDP dynamics.

Kantō and Kansai are two major economic regions, contributing 44.5% and 16.5% to GDP, respectively. The underestimation of GDP declines in Kantō and Kansai likely reflects omitted factors beyond the export shock. The Cabinet Office's annual report (the Japanese Economy and Public Finance 2008) identified declines in capital formation due to lower expected growth and in housing construction as significant shocks to the Japanese economy in 2008, second only to the export shock. Our model does not incorporate such medium- to long-term trends in capital formation and housing. Furthermore, fiscal stabilization efforts following the export shock likely had different effects across regions. Due to data limitations, it is challenging to attribute fiscal expenditures to specific regions with our current information, and therefore the investigation of this mechanism is left for future research.

Overall, Figure 10 shows that our model effectively captures regional variation in responses to export shocks, particularly in larger regions. This performance helps explain the national response shown in Figure 8, especially the early decline in GDP and consumption during the Great Recession.

7 Counterfactual experiments

Given that the model can account for a substantial share of the national decline in output and the regional heterogeneity in responses, it is worth examining the underlying mechanism. In particular, the mechanism for the inter-regional and inter-sectoral propagation of shocks is a distinctive feature of this model that warrants special attention.

In the following, we run counterfactual experiments to investigate the propagation mechanism in depth. In particular, we conduct a controlled experiment by feeding the model only the permanent shock to $y^f_{sj,t}$ in one region and industry sj at a specific t (2009Q1) and computing the new steady state, while keeping $y^f_{hi,t}$ for the other regions and industries $hi(\neq sj)$ constant. Here, we consider a negative export shock to the transportation equipment industry in the Chūbu region. We chose this industry and region because (i) the decline in automobile exports is one of the most important features of this recession and (ii) the headquarters of Toyota, the largest auto producer and exporter, is located in the Chūbu region.

7.1 Decomposition

To assess how an export-demand shock in one region affects other regions, we conduct a decomposition analysis based on distinct demand components. Our dynamic model includes four demand factors: domestic consumption, domestic investment, domestic intermediate-good demand, and foreign demand. In the following equation, the first term on the right-hand side represents domestic consumption demand; the second, investment demand; the third, domestic intermediate-good demand; and the fourth, foreign demand for good s produced in region j.

$$y_{sj,t} = \int_0^I (c^i_{sj,t} + x^i_{sj,t})di + \int_0^S \int_0^I m^{hi}_{sj,t} didh + y^f_{sj,t}. \tag{34}$$

Domestic consumption demand is given by

$$c^i_{sj,t} = \left(\frac{p_{sj,t}}{P^c_{i,t}}\right)^{-\sigma} \xi^i_{sjc} C_{i,t}. \tag{35}$$

Domestic investment demand is given by

$$x^i_{sj,t} = \left(\frac{p_{sj,t}}{P^x_{i,t}}\right)^{-\sigma} \xi^i_{sjx} X_{i,t}. \tag{36}$$

Domestic intermediate-good demand from industry h in region i is

$$m^{hi}_{sj,t} = \left(\frac{p_{sj,t}}{P^m_{hi,t}}\right)^{-\sigma} \gamma^{hi}_{sj} M_{hi,t}. \tag{37}$$

Given this demand structure, we compare two economies. The first is the baseline economy with no shocks. This economy remains at the 2008Q3 steady state. The second is the economy with an export shock in 2009Q1 that affects only one industry-region (the transportation equipment industry in the Chūbu region). We then hold the shock value constant at the 2009Q1 level and compute the new steady state. Comparing these two yields the overall changes in real sales in each region given this particular shock, where the real sales for region i are defined as follows:

$$\bar{Y}_{i,t} = \int_0^S p_{si,t=0} y_{si,t} ds. \tag{38}$$

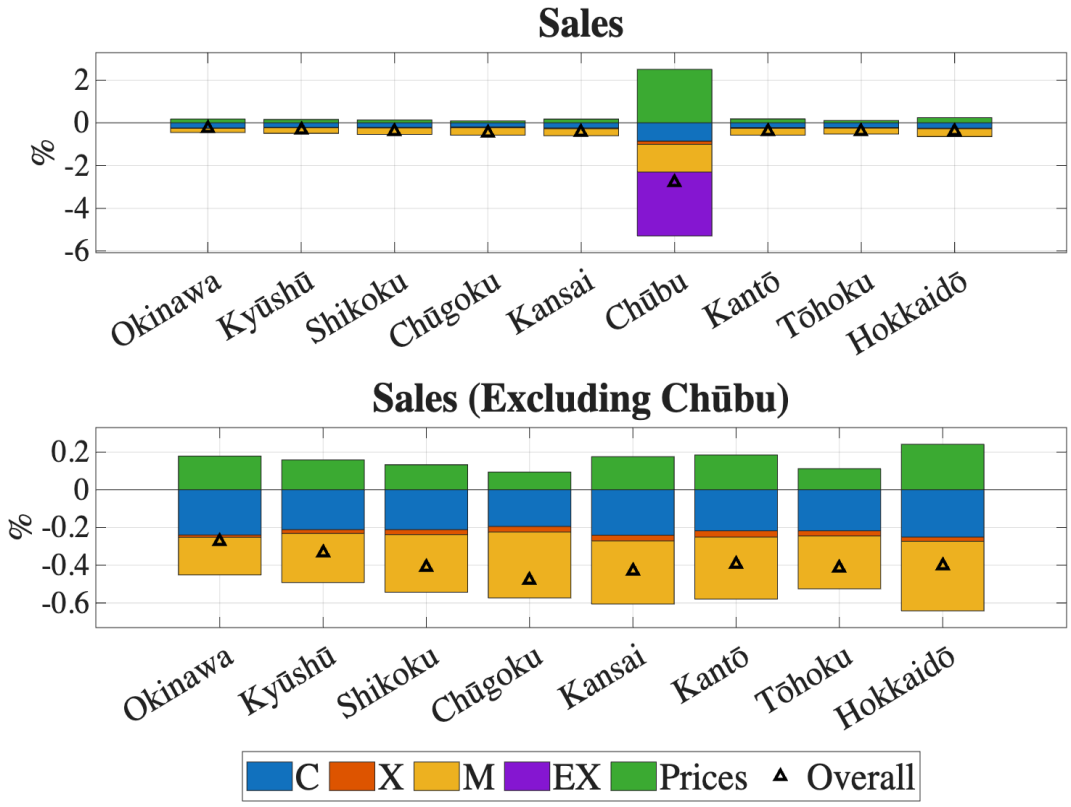


Figure 11: Changes in output following the 2009Q1 shock to the TE in Chūbu (2008Q3=1).

Note that we use the price at $t = 0$ to construct the real variable. Our decomposition exercise involves two steps. In the first step, we decompose the sales change in each region into five factors. The first factor is the effect of *prices*. These include $p_{sj,t}$ for all s and j and the price indices $P^c_{i,t}$, $P^x_{i,t}$, and $P^m_{hi,t}$ in (35), (36), and (37), which affect i's demand for goods produced in region j. Note that the foreign price $\bar{P}$ is fixed under the small open economy assumption. The second through fifth factors are $C_{i,t}$, $X_{i,t}$, $M_{hi,t}$, and $y^f_{sj,t}$. The first step identifies which factor affects a region's sales when each factor is aggregated across regions. The second step then decomposes each factor's contribution by the regions from which those effects originate.

Figure 11 presents the decomposition results. The figure illustrates how the region-industry-specific export shock propagates. The triangular dots show the overall effect of this shock on each region's total sales. The colored bars represent the decomposition. They are labeled as consumption C, investment X, intermediate goods M, exports y^f, and prices.

The first takeaway from the figure is that the effect of this shock on Chūbu itself is substantially larger than on the other regions that are not directly hit by the shock. At the same time, the propagation to the other regions is not negligible. The second takeaway is that, although the shock was on exports, the other demand components were also affected. The substantial drop in intermediate demand in Chūbu indicates that the intra-regional production network effect is important. Consumption in Chūbu also declines due to lower wages and profits. Because Chūbu's demand for goods falls, the price of these goods also falls, mitigating the overall effect on Chūbu's output.

To focus on inter-regional propagation, the bottom panel of Figure 11 plots the same objects

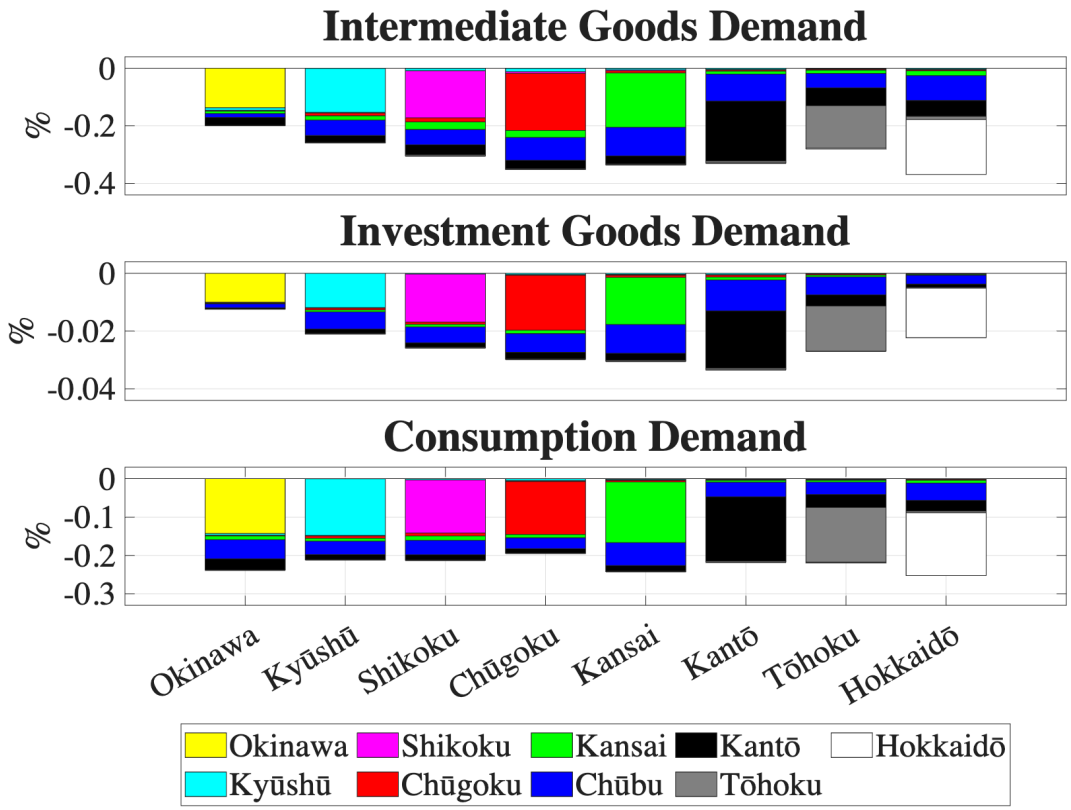


Figure 12: Changes in demand for intermediate, investment, and consumption goods following the 2009Q1 shock to the TE in Chūbu (2008Q3=1).

as the top panel but excludes the Chūbu region. The triangle dot shows the overall effect. All regions experience a reduction in overall sales from this shock to the Chūbu region's transportation equipment industry.

There are several takeaways about propagation. First, geography matters—the closer a region is to Chūbu, the more pronounced its overall decline tends to be. As shown in the map in Figure 4, the Chūbu region is roughly in the middle of Japan. Figure 11 lists the regions from west (Okinawa) to east (Hokkaidō). Thus, in the figure, the closer to the edges, the farther from the location of the shock. The black triangle shows that the decline is largest in the middle. This result is consistent with the gravity pattern of IRIO in Section 2.3.

Second, the negative effects are largely driven by consumption and intermediate-goods demand. The effect on investment goods is small. A novel finding here is the importance of consumption demand in the propagation process. In the macro-network literature, the focus has chiefly been on propagation through intermediate inputs. Our results show that the consumption linkage is quantitatively as important.

Third, the price effect, which attenuates the negative demand effect, is quantitatively significant. As in the Chūbu region, the price effect (along the demand curve) mitigates the impact of the demand decline. This result underscores the importance of modeling the general equilibrium properly. Section 7.2 below examines the impact of price flexibility in more detail.

In the second step of the decomposition, we examine each demand factor—intermediate input, investment, and consumption—separately and decompose the change into its regional components.

Figure 12 shows each region's contribution to the column region's decline in intermediate input, investment, and consumption demand, respectively.

Two components stand out as quantitatively important across all three graphs. First, the decline in demand from Chūbu, where the shock hits, is substantial. The shock to Chūbu reduces demand for intermediate goods through the input-output network. It also reduces demand for consumption and investment goods from other regions because of the decline in Chūbu consumers' income. This cross-regional demand propagation, in turn, reduces income in other regions. Second, this decline in income in regions other than Chūbu reduces demand for goods from the home region. This secondary effect turns out to be quantitatively more important than the direct effect from the Chūbu region. This suggests a regional multiplier effect—reduced demand from other regions lowers demand from the home region—as we examine in detail in Section 7.3.

In the top panel of Figure 12, we observe a mild gravity pattern. Recall that Chūbu is located in central Japan (see Figure 4) between Kansai and Kantō, with regions ordered from west to east. Focusing on the Chūbu effect, shown in blue, we see that negative effects are strongest in the central regions and weaken toward the edges, except in Hokkaidō. This again aligns with the gravity structure identified in the IRIO matrix (Section 2.3). The overall demand decline, including effects from regions other than Chūbu, follows a similar pattern. We see a more prominent gravity pattern for investment goods (middle panel of Figure 12) and no gravity pattern for consumption goods (bottom panel of Figure 12). The heterogeneity of spatial propagation across channels is a novel finding of this paper.

7.2 The role of price flexibility

The model above is in the real business cycle tradition in that all prices are flexible. In the decomposition, we see that price flexibility does indeed play a role: the effect of price changes mitigates the output decline associated with negative demand shocks. In this section, to further investigate how prices affect propagation, we make the opposite assumption: we fix all prices at the $t = 0$ level.

Figure 13 repeats Figure 11 under fixed prices. The main difference for the Chūbu region is the absence of the price effect, which amplifies the negative impact on output. The composition of each component is almost identical.

Figure 14 plots the results corresponding to Figure 12 shown earlier. These figures look almost identical to those for flexible prices, except that the magnitude of the negative effects is larger than in the flexible-price setup.

7.3 Regional multiplier

The only source of shocks in our model is foreign export demand. Thus, if a region experiences no export shock, it is affected only by transactions with other regions. When the region is sufficiently small, we can neglect the national equilibrium effect, in which the region's response to other regions' demand creates ripple effects through its demand for other regions' output. Then, we can estimate the multiplier as the ratio of the region's GDP increase to the increase in demand from other regions for the region's output.

Under this "small-region" assumption, the multiplier formula (25) applies to a region's GDP response to an increase in inter-regional inbound demand, with TS reinterpreted as a region's share of inter-regional trade. In our baseline calibration, the average inter-regional trade share

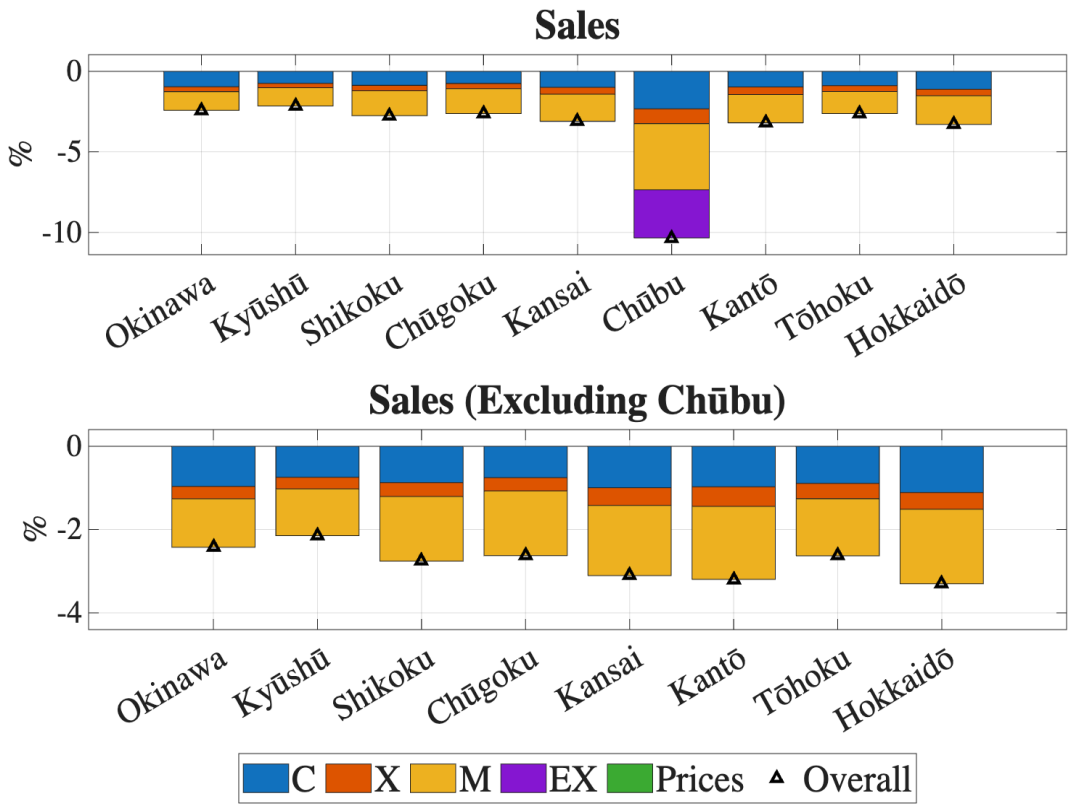


Figure 13: Changes in output following the 2009Q1 shock to the TE in Chūbu (2008Q3=1, fixed prices).

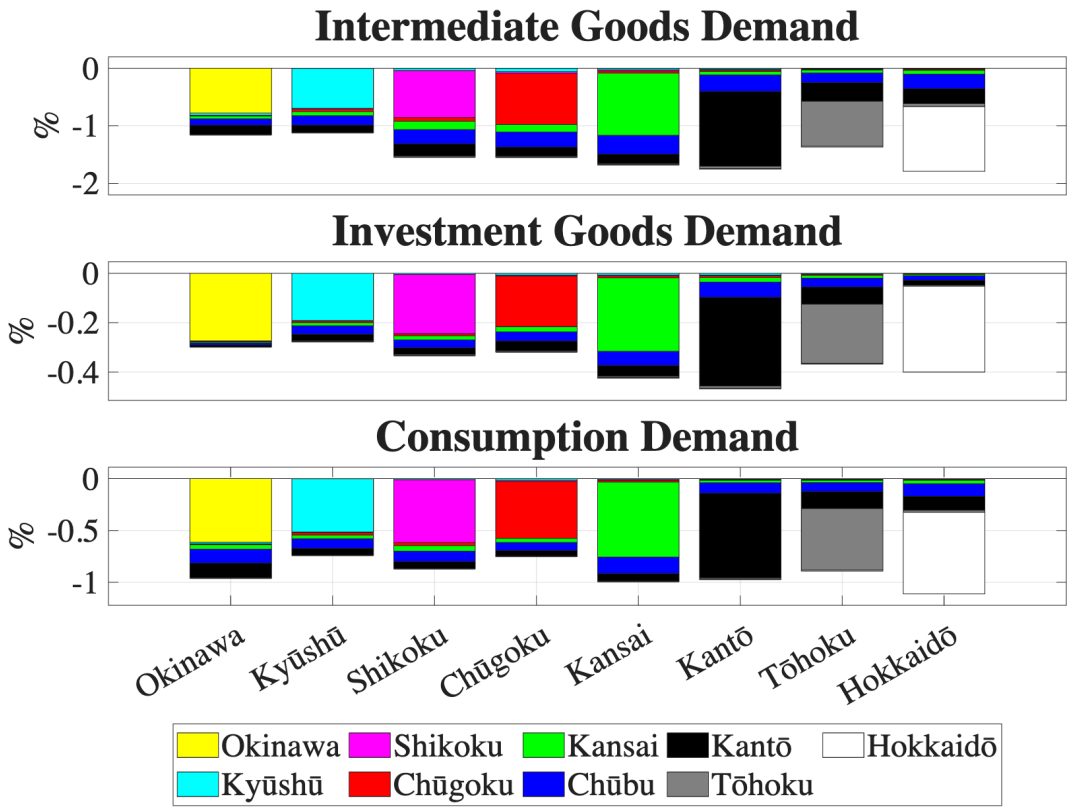


Figure 14: Changes in demand for intermediate, investment, and consumption goods following the 2009Q1 shock to the TE in Chūbu (2008Q3=1, fixed prices).

of regional GDP is 22.2%, ranging from 13.2% for Kantō to 27.5% for Shikoku. Using (28), the regional multiplier averages 1.15, with a maximum of 1.72 for Kantō and a minimum of 0.94 for Shikoku. Hence, a one-yen increase in inbound demand from outside raises regional GDP by more than one yen on average, with the effect larger for larger regions with a higher share of home demand.

This analysis clarifies a propagation-and-amplification mechanism in our model. First, an export shock hits a region and generates a multiplier effect within that region. The multiplier formula captures this direct impact, ignoring inter-regional trade. For example, if an export shock hits Chūbu, whose export share of sales is 8.4%, (28) implies a multiplier of 1.62. Chūbu's income declines, triggering an inter-regional demand shock in other regions, which have an average multiplier of 1.15 within those regions. Hence, the export shock on Chūbu amplifies and propagates to other regions with multipliers greater than one, generating a national-level multiplier effect.

The direct income impact is mitigated by price effects, in which Chūbu's price falls and draws demand from other regions. This means that other regions experience an inter-regional "terms-of-trade" shock against Chūbu. This effect is not captured by the multiplier formula because the outside price (p_f) is fixed in the formula.

To estimate the regional multiplier in the full model, we use our counterfactual experiment in which Chūbu's transportation equipment (TE) export demand ($y^f_{\text{Chūbu, TE}}$) declines exogenously by 10%. We calculate equilibrium changes in region i's GDP and in intermediate and final demands for i from all other regions. The estimated regional multiplier averages 2.30, ranging from 2.05 in Chūgoku to 2.55 in Kantō. We note that this estimate exceeds the value obtained from (28). This is because the price fall in Chūbu exerts a business-stealing effect in addition to the effect of decreased demand from Chūbu. At the same time, it partially offsets Chūbu's sales loss. Hence, price effects propagate the shock across regions but do not necessarily amplify the national-level response.

These estimates align with Nakamura and Steinsson (2014), who estimated the relative multiplier effects of military procurement spending on regional GDP to range from 1.4 to 1.9 and 2.5 to 2.8, depending on the estimation methods and data. In our full model, the relative multiplier (i.e., the GDP response to a marginal variation in inter-regional inbound demand) is 2.53, which differs little from the average multiplier.

Using the same experiment, we also compute a national-level multiplier, defined as the decrease in GDP divided by the exogenous 10% decline in Chūbu's TE, which equals 1.82. We note that this multiplier is broadly consistent with the size of the impulse and response observed in Japan in 2008—the ratio of GDP decline to export decline, 1.43. Moreover, the multiplier computed by the full model is 32% larger than the multiplier derived from the multiplier formula (28), 1.38. This suggests that the heterogeneous disaggregated model has only a modest amplification effect compared to the symmetric aggregate model. We will discuss this observation in Section 8.2.

Our model provides a structural explanation of the multiplier effect. In the model, increased demand from other regions raises labor demand and output among local firms, leading to higher equilibrium wages and longer hours worked. The resulting rise in labor income then boosts regional demand, particularly in the service sector, which primarily relies on local consumers. Notable exceptions include tourism, inter-regional distribution margins, and intra-firm trade in managerial services, all of which are explicitly accounted for in our IRIO tables.

The share of the non-service sector varies across regions from 12.0% in Okinawa to 48.9% in Chūbu, averaging 32.2%. Even within tradable sectors, 51.3% of demand originates in the home

region. This strong home bias is also reflected in our gravity estimates (Section 2.3). Therefore, the home bias in goods demand underpins the regional multiplier effect in our model, just as the export share determines the export multiplier in Proposition 1(b). This mechanism is akin to the local multiplier in the model of the government spending multiplier at the county level developed by Dupor et al. (2023).

8 The static model, once again

Finally, we reassess the static model used to obtain an analytical solution in Section 4 to further explore how shocks propagate in the model. Unlike Section 4, we do not impose symmetry across sectors and regions; the model remains quantitatively evaluable. This section makes two novel contributions.

First, we analytically derive the explicit formula for the decomposition conducted in Section 7. The formula links the decomposition to the model's fundamental forces. Because the simplifying assumptions are minimal, we can still evaluate the outcome of the counterfactual experiment quantitatively.

Second, we use the analytical formula, which approximates local responses, to assess the importance of nonlinearity when analyzing responses to large shocks. The comparison is between the local (small-shock) response from the analytical formula and the full static model (presented in Appendix F). We find that the model's nonlinearity is quantitatively important, and thus the local approximation may yield misleading results in the case of large shocks.

8.1 First-order effects of export shocks in a static model with $\sigma_c = 1$ and $\gamma^{\cdot}_{\cdot f} = 0$

We begin with the static version of our quantitative model in Section 3. The details of the static model are given in Appendix F. The only differences from the model in Section 3 are that we remove the saving decision from the consumer's problem and exclude capital stock as a factor of production.

Here, we impose minimal simplifying assumptions on the static model: $\sigma_c = 1$, $\gamma^i_{sf} = 0$, and $\alpha_s = \alpha$; that is, labor supply is inelastic (holding the trade balance fixed), imports are not used as intermediate inputs, and α_s is homogeneous across industries. With these assumptions, we derive an analytical expression for the comparative statics of ω^f below without sacrificing heterogeneity across regions and sectors (i.e., maintaining the full input-output structure $(\xi^i_{sj}, \gamma^{hi}_{sj})$). We present only the main result and defer the detailed derivations to Appendix E.

First, we introduce the notation used below. Let $q_{si} := (p_{si})^{\sigma-1}$, $Q^{si} := (P^{si})^{\sigma-1}$, and $Q^i := (P^i)^{\sigma-1}$ denote prices, and $Y^{si} := p_{si}y_{si}$ denote sales. Let $\boldsymbol{q}$ be an $SI \times 1$ vector formed from the matrix $[q_{si}]$. Define $\boldsymbol{Q}$, $\boldsymbol{Y}$, and $\boldsymbol{\omega}^f$ similarly. We let $Y^{0i} := \sum_{s=1}^{S} Y^{si}$ and denote its vectorized form by $\boldsymbol{Y}^0$. The matrices $\mathrm{diag}\,(qY)$, $\mathrm{diag}\,(Q)$, $\mathrm{diag}\,(\omega^f)$, and Γ_0 are constructed from these vectors.[14] We denote international transfers as a share of regional sales by $b^i := B_i/Y^{0i}$, and $\boldsymbol{b}_I$ is its length-I vector.

With this notation, we obtain the following decomposition of the equilibrium response of sales to small perturbations in $d\ln\boldsymbol{\omega^f}$ and $d\boldsymbol{b}_I$, representing shocks to exogenous export demand and

[14] $\mathrm{diag}\,(qY)$ denotes a diagonal matrix with $q_{si}Y^{si}$ in the $(S(i-1)+s)$-th element. Similarly, $\mathrm{diag}\,(Q)$ and $\mathrm{diag}\,(\omega^f)$ denote diagonal matrices with Q^{si} and ω^f_{si}, respectively. Γ^0 is an $SI \times I$ matrix containing $\hat{\gamma}^{0j}_{si}Q^{0j}Y^{0j}$ in $(S(i-1)+s, j)$.

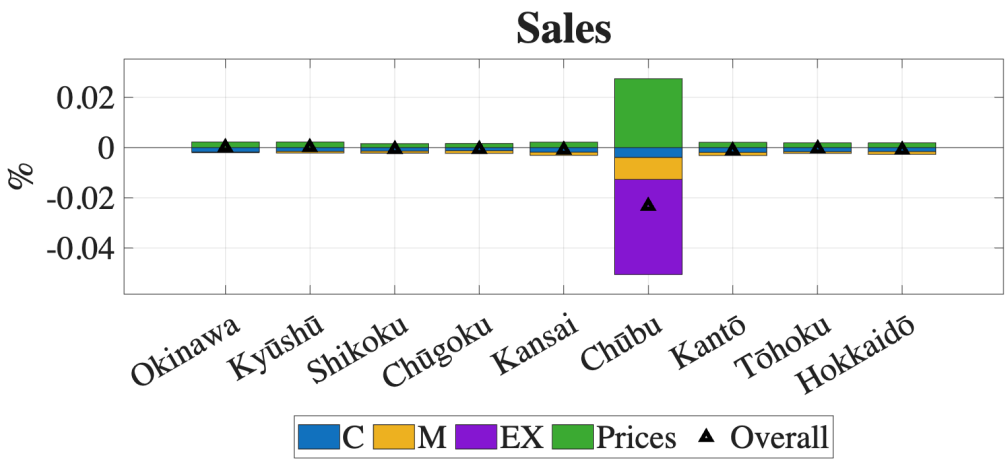


Figure 15: Analytical decomposition of an export demand shock on regional outputs. One percent negative shock hits the Transportation Equipment sector in the Chūbu region.

international transfers.

$$d\ln \boldsymbol{Y} = \operatorname{diag}(qY)^{-1}\Bigg[\underbrace{\Gamma^0 \operatorname{diag}\left(\frac{1}{1-\alpha(\sigma-1)/\sigma+b_I}\right)d\boldsymbol{b}_I + \operatorname{diag}(\omega^f)(\bar{P})^{\sigma} d\ln \boldsymbol{\omega^f}}_{\text{direct exogenous effects}}$$
$$+\underbrace{Gd\ln \boldsymbol{Y}}_{M\text{ effect}}+\underbrace{\Gamma^0 d\ln \boldsymbol{Y}^0}_{C\text{ effect}}+\underbrace{Gd\ln \boldsymbol{Q}+\Gamma^0 d\ln \boldsymbol{Q}^0-\operatorname{diag}(qY)d\ln \boldsymbol{q}}_{\text{Price effects}}\Bigg] \tag{39}$$

An explicit solution for prices $d\ln \boldsymbol{q}$ as a linear function of the exogenous shocks is provided in Appendix E. The analytical decomposition facilitates interpretation of the comparative statics. The second line of (39) shows how export shocks propagate through intermediate goods demand M, consumption goods demand C, and price adjustments. The M effect corresponds to the traditional input-output propagation of demand quantities, whereas the price effects capture substitution channels. The C effect represents the general equilibrium effect, in which changes in output affect final demand through regional income.

The question we ask below is whether we can use this local decomposition formula to analyze the propagation of shocks, as we did in Section 7. The answer is twofold. First, for small shocks, the formula yields a quantitative outcome similar to that of the full model. Second, for large shocks, such as the export shock analyzed in this paper, scaling the local response deviates significantly from the full nonlinear solution.

For the first point, Figure 15 illustrates a decomposition exercise in which the export demand for transportation equipment in the Chūbu region falls by one percent.[15] Here, we impose a small shock to express the local approximation. Because we set $\gamma^i_{sf} = 0$ in this exercise, we recalibrate the

[15] We assume that the regional international transfer b^i is zero in the initial equilibrium and shifts one-to-one with changes in the region's net exports.

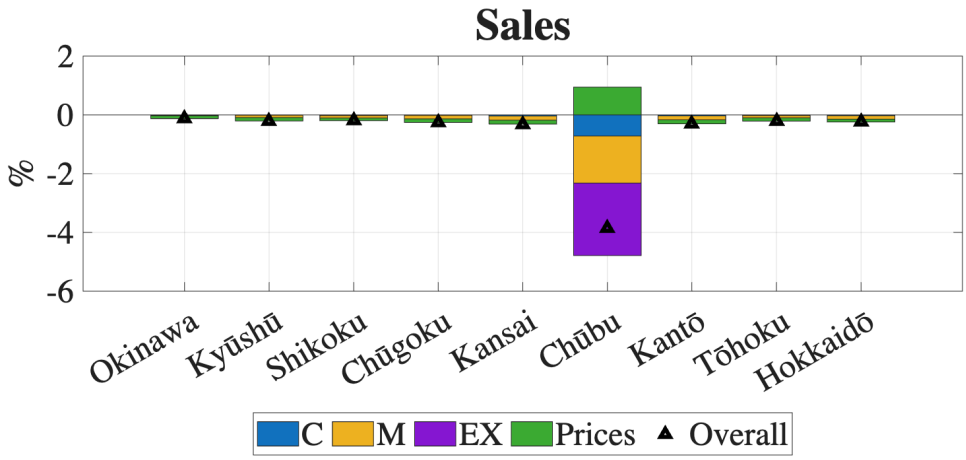


Figure 16: Changes in output with 2009Q1's shock to the TE in Chūbu (2008Q3=1).

static model presented in Appendix F accordingly and use the resulting parameters to implement the decomposition in (39).

The decomposition shows three features we already saw in Section 7 (Figure 11). First, the region affected by the export shock (i.e., Chūbu) experiences a significant decline in output, and the shock also reduces output in other regions. Second, propagation through consumption is as important as propagation through intermediate demand. Third, price effects are substantial and largely offset the demand propagation effects in regions outside Chūbu. Thus, the locally approximated decomposition provides a reasonable approximation of the full model for small shocks.

For the second point, Figure 16 presents the same decomposition, computed numerically with a static model that does not rely on the analytical formula for the first-order effect. We use the recalibrated version of the model with $\gamma_{sf}^{i} = 0$ to ensure comparability with the results based on (39). Here, we impose a large shock: the reduction in the transportation equipment sector in the Chūbu region is 62.8%, matching the observed value in 2009Q1.

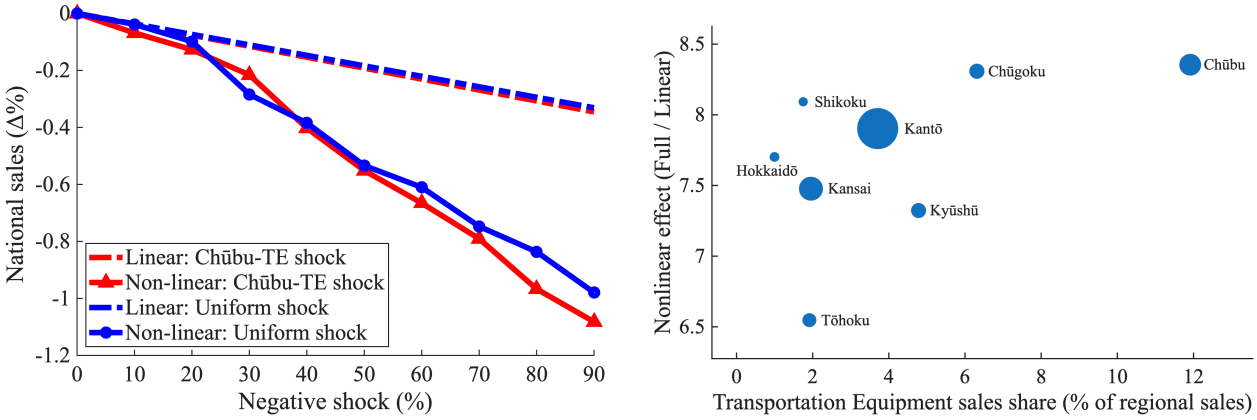
We find that (i) extrapolating the local effect (that is, multiplying the result in Figure 15 by 62.8) would underestimate propagation, (ii) this underestimation is larger for non-Chūbu regions than for the Chūbu region, and (iii) the mitigating effects of price adjustments outside Chūbu are evident only in the local approximation (Figure 15) and not in the full model (Figure 16).

The experiment above indicates the presence of important nonlinearity embedded in the model. In the next section, we further explore the difference between the local (linear) approximation and the full nonlinear solution.

8.2 Nonlinearity

To further investigate the model's nonlinearity, we compare first-order effects with those from the full model. To facilitate the comparison, we use the static version of the model (see Appendix F for the model setup). More specifically, we consider export shocks of varying magnitudes to the transportation equipment (TE) industry in Chūbu and compute the resulting changes in national sales. This exercise shows how much negative export shocks to a particular region propagate to

Figure 17: Nonlinearity of shock propagation



(a) Linear and full effects by shock size

(b) Nonlinearity by shock origin

Note: Panel (a) compares the effects implied by the full model and the first-order approximation on national sales following negative export-demand shocks of varying magnitudes to the transportation equipment (TE) industry in Chūbu. The line with circular markers shows the case in which shocks hit all region-industry pairs uniformly. Panel (b) reports the degree of nonlinearity across shock-origin regions, defined as the ratio of the decline in the sum of sales in regions $j \neq i$ implied by the full model to that implied by the first-order approximation, following a 50% decline in export demand for the TE industry in region i. Marker size is proportional to each region's share of national sales.

other regions and how well the first-order approximation captures the full-model effects. Figure 17 (a) presents the results.

We find that the nonlinearity is substantial. The gap between the full model and the first-order approximation widens—in both absolute and relative terms—as the shock size increases. For small negative shocks, the two approaches yield nearly identical effects, indicating that the full-model response is approximately linear over this range. At a 50% decline in export demand, similar in magnitude to the observed shock, the full model predicts a -0.54% decline in national sales, whereas the first-order approximation implies only a -0.18% decline. The full-model effect is about three times as large as the first-order effect.

Inspections of the model outcomes show that the nonlinear effects capture nonlinear wage responses. With our setting of $\sigma_c = 1$, the regional labor supply response is rigid. Therefore, a negative shock to inter-regional demand shifts labor demand down and reduces equilibrium wages, with the magnitude of the response determined by the elasticity of labor substitution across industries, τ. Low labor income in the tradable goods sector then spreads to the non-tradable goods sector in the region, causing a regional multiplier effect, as discussed in Section 7.3.

The nonlinear effect is pervasive across regions. Figure 17 (b) plots the degree of nonlinearity, defined as the ratio of the decline in the sum of sales in regions $j \neq i$ implied by the full model to that implied by the first-order approximation, following a 50% decline in export demand for the TE industry in region i. The horizontal axis reports the TE industry's regional sales share. Two findings emerge. First, the degree of nonlinearity is positively related to the regional importance of the industry hit by the shock, as measured by its share of regional sales. Second, the nonlinear effect is substantial regardless of the regional origin of the shock. For example, even in Tōhoku, where the TE industry accounts for only about 2% of regional sales, the full model predicts an effect more than six times larger than the first-order approximation.

The nonlinearity we find does not stem from heterogeneity per se. To see this, we conduct an

experiment using a uniform shock in place of the Chūbu-TE shock. The uniform shock affects the export demand of all region-industries by the same proportion, while keeping the decline in total exports unchanged relative to the Chūbu-TE shock. The result is shown in Figure 17 (a). The impact of the uniform shock deviates little from that of the Chūbu-TE-specific shock.

The uniform experiment yields three findings. First, the shock heterogeneity generates little amplification relative to an aggregate or uniform shock, consistent with Section 7.3 and with a sectoral real business cycles analysis by Dupor (1999). Second, flexible prices play a key role in mitigating the region-sectoral shock. Because wage responses are nonlinear in each region, the region hit by a negative shock experiences a large wage decline. However, increased demand from other regions mitigates the direct impact. The irrelevance of sectoral shocks suggests that the price mechanism spreads the region-industry-specific shock so that aggregate income remains at the level of the uniform shock case. Third, the locally linearized sectoral model fails to approximate the aggregate response even when the shock is uniform. This contrasts with the linearized model's ability to reproduce the propagation pattern shown in Section 8.1, in which the region hit by the export-demand shock bears a disproportionately large impact, while other regions experience smaller effects that depend on their input-output linkages with the affected region. This suggests that the higher-order effects of a sectoral model constitute much of the aggregate response captured by the nonlinear solution of the full model or by the first-order effect (25) of the aggregate model.

9 Conclusion

This study develops a multi-region, multi-sector model to analyze how export shocks propagate in Japan during the Great Recession. Our model features monopolistic competition, inter-regional input-output linkages, and a representative consumer in each region.

We measure export shocks in each region using trade statistics. The inter-regional input-output matrix, unique to Japan, allows us to analyze how shocks propagate through the input-output network. Our model also explicitly accounts for consumers, whose final goods demand is affected by shocks through changes in income.

Calibrating the model to 2008Q3, we examine how the model's response to export shocks compares with the data. We find that the model with flexible prices can replicate about three-fourths of the output decline and the full consumption decline at the macro level. At the regional level, the export shock has a particularly large impact on output in regions where exports account for a large share of regional GDP.

We run counterfactual experiments to examine how shocks propagate across regions and industries. In the main experiment, we feed the model an export shock that affects only a particular industry (transportation equipment) in a specific region (Chūbu). We find that a shock to one region and industry propagates to other regions through consumption demand and IO linkages. The effect is especially pronounced for geographically closer regions. The secondary effect of the decline in own consumption caused by the income drop is also important. We find that the decline in prices substantially attenuates the negative effects. An experiment without flexible prices confirms this point: both within- and across-region output declines are significantly larger under fixed prices.

We analyze an aggregatable symmetric case to illustrate how an export demand shock translates into a terms-of-trade shock under a monopolistic competition structure. The simple model yields a multiplier formula for an export demand shock, which tends to be large when the export share is

small. The simple formula matches the GDP response to the export shock in Japan in 2008 well. Our disaggregate model further explains the heterogeneous responses across regions and industries.

The analytical version of our disaggregated model identifies the channels through which a shock is transmitted. A comparison between the local approximation of the simple model and the full nonlinear solution indicates that the model's propagation mechanism exhibits significant nonlinearity, and that the local linear approximation of the disaggregated model would underestimate the magnitude of transmission, especially for large shocks.

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Appendix

A Mapping prefectures into regions

Table 3 below describes the correspondence between prefectures and the regions we use in the study.

Regions	Prefectures
Hokkaidō	Hokkaidō
Tōhoku	Aomori, Iwate, Miyagi, Akita, Yamagata, Fukushima
Kantō	Ibaraki, Tochigi, Gunma, Saitama, Chiba, Tōkyō, Kanagawa, Niigata, Yamanashi, Nagano, Shizuoka
Chūbu	Toyama, Ishikawa, Gifu, Aichi, Mie
Kansai	Fukui, Shiga, Kyōto, Ōsaka, Hyōgo, Nara, Wakayama
Chūgoku	Tottori, Shimane, Okayama, Hiroshima, Yamaguchi
Shikoku	Tokushima, Kagawa, Ehime, Kōchi
Kyūshū	Fukuoka, Saga, Nagasaki, Kumamoto, Ōita, Miyazaki, Kagoshima
Okinawa	Okinawa

Table 3: Region classification in our model. This classification is based on the inter-regional input-output data (2005) provided by the Ministry of Economy, Trade and Industry (METI).

B Details of the data construction

We use the Trade Statistics of Japan (TSJ), compiled by the Ministry of Finance (MoF), to construct a quarterly series of exports for each industry, sj.[16] The TSJ monthly reports the values of 28 exported goods at ten customs, which can be further broken down into 166 offices. We map each office to our regional classification and each good to our industry classification, then aggregate the raw data to construct the quarterly export series for each region-industry.

Two points should be noted. First, the TSJ does not capture trade in services. In the counterfactual experiments, where we feed the measured export shocks into each industry, we therefore assume that service industries—such as "Services for Consumers" and "Services for Businesses"—follow the same export path as "Misc" (miscellaneous) in Figure 6, in terms of the export decline relative to the benchmark level. Note that service trade is observed in the IRIO at a single point in time (2005), which pins down the level of service exports in the initial economy; only the subsequent path is therefore assumed to follow that of "Misc." Second, because the TSJ does not cover all industries, the implied aggregate exports do not match those reported in the SNA. When computing the demand parameters $\omega^f_{sj,t}$ for each sector sj and period t, we therefore use the TSJ data to replicate the export shares of each sector sj—that is, the share of exports of good sj in total exports—in each period, while scaling them so that aggregate exports match those reported in the SNA.

[16] See https://www.customs.go.jp/toukei/info/index_e.htm.

C More on Calibration

(i) Static model

#	Industry (our classification)	JIP(2008)	A_s	α_s
1	Agriculture, Forestry, Fisheries	1-6	1.000	0.71
2	Mining and Quarrying of Stone and Gravel	7	0.497	0.72
3	Food and Beverage	8-14	2.596	0.79
4	Textile Mill Products	15	2.078	0.66
5	Pulp, Paper, and Paper Products	18	1.631	0.82
6	Chemical Products	23-29	1.975	0.86
7	Petroleum and Coal Products	30,31	3.149	0.98
8	Ceramic, Stone, and Clay Products	32-35	0.813	0.66
9	Iron and Steel	36,37	3.412	0.90
10	Non-Ferrous Metals	38,39	1.271	0.83
11	Fabricated Metal Products	40,41	2.154	0.63
12	General-Purpose Machinery	42-45	2.879	0.72
13	Electrical Machinery	46-53	3.165	0.75
14	Transportation Equipment	54-56	5.876	0.82
15	Information and Communication Electronics Equipment	57	1.188	0.67
16	Miscellaneous Manufacturing Products	16,17,19-22,58,59	2.092	0.69
17	Construction	60,61,72	15.465	0.77
18	Electricity, Gas, Heat Supply and Water	62-66	4.330	0.81
19	Wholesale and Retail Trade	67,68	18.296	0.50
20	Finance, Insurance, and Real Estate	69-71	8.289	0.50
21	Transportation	73-77	4.350	0.59
22	Information and Communication	78,79, 90-93	4.308	0.67
23	Education, Medical, Health Care, and Welfare	80-84,98-107	7.578	0.39
24	Services for Businesses	85-88,	5.754	0.60
25	Services for Consumers	89,94-97	4.802	0.57
26	Others	108	1.389	0.98

Table 4: Industry classification for our model, its correspondence with the JIP Database, factor-neutral productivity (with agriculture=1), and intermediate goods share. These parameters are computed using the JIP Database 2008 (https://www.rieti.go.jp/en/database/JIP2008/index.html), provided by the RIETI. We map the JIP Database's industry classification to ours and compute the industrial TFP and intermediate shares.

	Okinawa	Kyushu	Shikoku	Chugoku	Kansai	Chubu	Kanto	Tohoku	Hokkaido
A_i	0.978	1.077	1.172	1.253	1.306	1.348	1.377	1.030	1.000
χ_i	8.8e-7	4.7e-11	1.2e-13	2.5e-14	3.8e-12	3.5e-19	6.5e-9	1.4e-13	1.2e-10

Table 5: Parameter values for the regional TFP and disutility of labor

Tables 4 and 5 list parameter values used to calibrate the static model. Figures 18a, 18b, and 18c are heatmaps of additional parameter values.

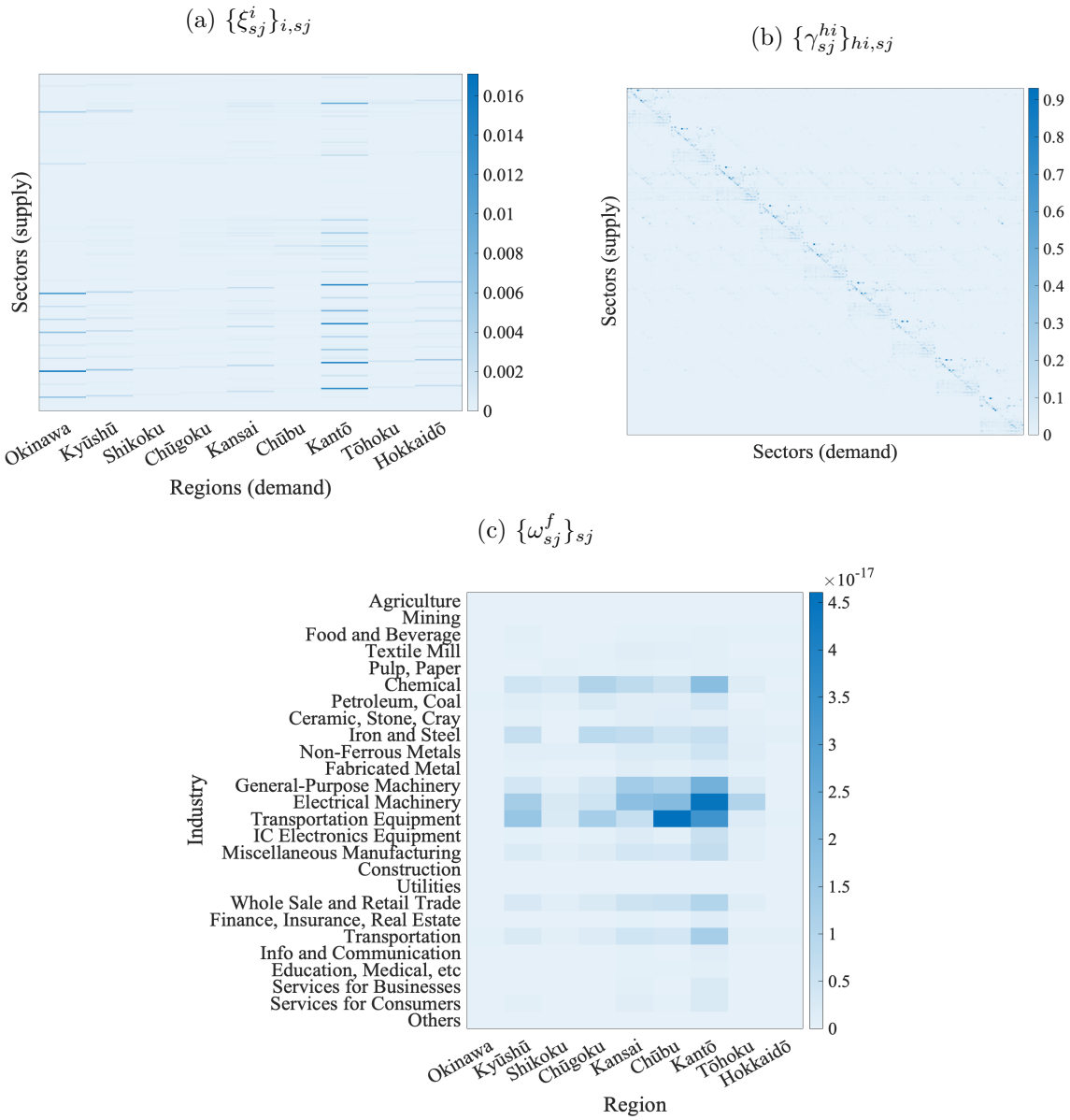


Figure 18: Values for the weight parameters

(ii) Dynamic model

#	Industry (our classification)	JIP(2008)	A_s	α_s	β_s
1	Agriculture, Forestry, Fisheries	1-6	1.000	0.51	0.21
2	Mining and Quarrying of Stone and Gravel	7	0.497	0.63	0.24
3	Food and Beverage	8-14	2.596	0.72	0.19
4	Textile Mill Products	15	2.078	0.59	0.30
5	Pulp, Paper, and Paper Products	18	1.631	0.74	0.16
6	Chemical Products	23-29	1.975	0.77	0.13
7	Petroleum and Coal Products	30,31	3.149	0.94	0.02
8	Ceramic, Stone, and Clay Products	32-35	0.813	0.58	0.29
9	Iron and Steel	36,37	3.412	0.82	0.09
10	Non-Ferrous Metals	38,39	1.271	0.76	0.16
11	Fabricated Metal Products	40,41	2.154	0.59	0.35
12	General-Purpose Machinery	42-45	2.879	0.65	0.25
13	Electrical Machinery	46-53	3.165	0.66	0.22
14	Transportation Equipment	54-56	5.876	0.75	0.16
15	Information and Communication Electronics Equipment	57	1.188	0.55	0.28
16	Miscellaneous Manufacturing Products	16,17,19-22,58,59	2.092	0.63	0.28
17	Construction	60,61,72	15.465	0.44	0.22
18	Electricity, Gas, Heat Supply and Water	62-66	4.330	0.54	0.14
19	Wholesale and Retail Trade	67,68	18.296	0.46	0.45
20	Finance, Insurance, and Real Estate	69-71	8.289	0.42	0.40
21	Transportation	73-77	4.350	0.47	0.34
22	Information and Communication	78,79, 90-93	4.308	0.56	0.29
23	Education, Medical, Health Care, and Welfare	80-84,98-107	7.578	0.33	0.52
24	Services for Businesses	85-88,	5.754	0.46	0.34
25	Services for Consumers	89,94-97	4.802	0.50	0.38
26	Others	108	1.389	0.98	0.02

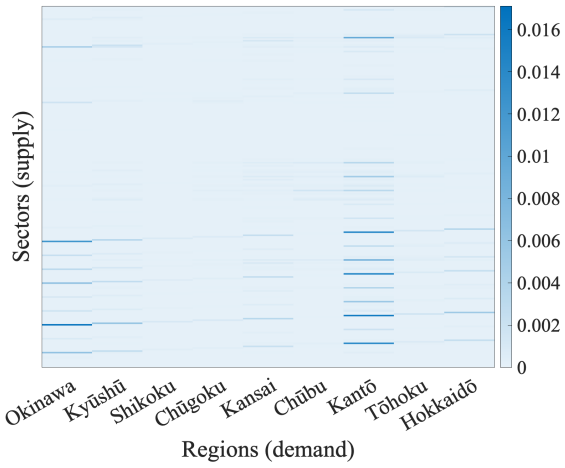
Table 6: Industry classification for our model, its correspondence with the JIP Database, factor-neutral productivity (with agriculture=1), and intermediate goods share. These parameters are computed using the JIP Database 2008 (https://www.rieti.go.jp/en/database/JIP2008/index.html), provided by the RIETI. We map the JIP Database's industry classification to ours and compute the industrial TFP and intermediate shares.

	Okinawa	Kyūshū	Shikoku	Chūgoku	Kansai	Chūbu	Kantō	Tōhoku	Hokkaidō
A_i	0.978	1.077	1.172	1.253	1.306	1.348	1.377	1.030	1.000
χ_i	35212.8	12.8	855.2	89.1	2.2	9.1	8.2	39.6	262.3

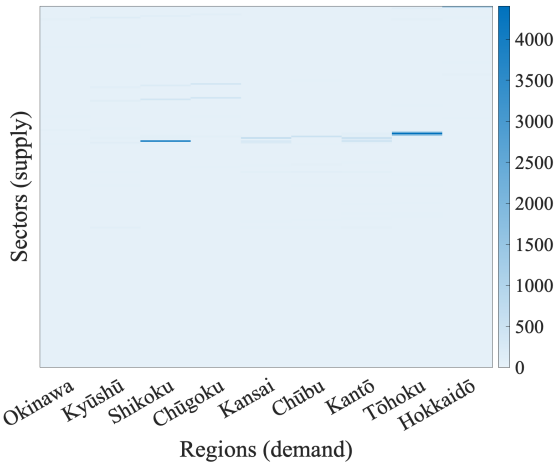
Table 7: Parameter values for the regional TFP and disutility of labor.

Tables 6 and 7 list parameter values used to calibrate the baseline dynamic model. Figures 19a, 19b, 19c, and 19d are heatmaps of additional parameter values.

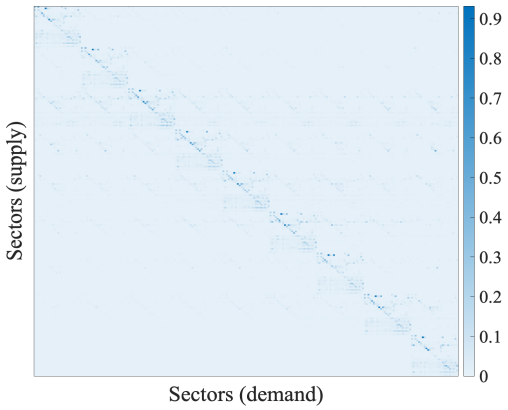
(a) $\{\xi^i_{sjc}\}_{i,sj}$



(b) $\{\xi^i_{sjx}\}_{i,sj}$



(c) $\{\gamma^{hi}_{sj}\}_{hi,sj}$



(d) $\{\omega^f_{sj}\}_{sj}$

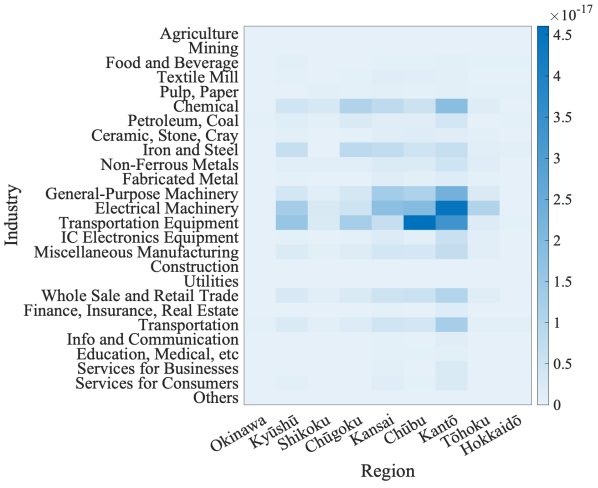


Figure 19: Values for the weight parameters.

Details on the calibration procedure: To determine the distribution of transfers across regions, we assume that each region's share of transfers equals its share of national GDP:

$$\frac{B_{i,0}}{\int_0^I B_{j,0}\, dj} = \frac{GDP_{i,0}}{\int_0^I GDP_{j,0}\, dj},$$

where $GDP_{j,0}$ denotes the regional GDP of region j in the initial economy. Because the calibration of $\{\omega^f_{si}\}_{si}$ ensures that the model's national export-to-GDP ratio matches the data, replicating net exports implies that the import-to-GDP ratio also matches the data.

Although it would be ideal to determine the weight parameters for imported goods ($\{\xi^i_{sfc}\}_{i,s}$, $\{\xi^i_{sfx}\}_{i,s}$, and $\{\gamma^{hi}_{sf}\}_{hi,s}$) using the same procedure as for their domestic counterparts (i.e., the strategy used to pin down $\{\xi^i_{sj}\}_{i,sj}$ and $\{\gamma^{hi}_{sj}\}_{hi,sj}$), the corresponding target moments are unavailable due to data limitations. In particular, there is no information on (i) the share of goods $s \in S$ that region i imports from foreign countries, and (ii) how such goods are allocated across consumption, investment, and intermediate use. Therefore, we assume that each region i allocates imported goods in the same proportions as domestically produced goods. Specifically, given the total imports of region i, the share of good s from foreign countries allocated to consumption equals the share of domestically produced good s in total domestic consumption. The same assumption applies to investment and intermediate use. We formalize these assumptions below.

Let IM_i denote the total imports for region i:

$$IM_i = \int_0^S p_{sf,t}\left(c^i_{sf,t} + x^i_{sf,t} + \int_0^S m^{hi}_{sf,t} dh\right) ds.$$

Likewise, let DD_i denote the total domestic demand for region i:

$$DD_i = \int_0^I \int_0^S p_{sj,t}\left(c^i_{sj,t} + x^i_{sj,t} + \int_0^S m^{hi}_{sj,t} dh\right) dsdj.$$

Our assumption implies that

$$\frac{p_{sf,t}c^i_{sf,t}}{IM_i} = \frac{\int_0^I p_{sj,t}c^i_{sj,t}dj}{DD_i} \quad \forall (i,s) \in I \times S$$

and

$$\frac{p_{sf,t}x^i_{sf,t}}{IM_i} = \frac{\int_0^I p_{sj,t}x^i_{sj,t}dj}{DD_i} \quad \forall (i,s) \in I \times S.$$

Finally, let M^d_i and M^f_i denote region i's total expenditures on intermediate goods produced in Japan and in foreign countries:

$$M^d_i = \int_0^I \int_0^S p_{sj,t} \int_0^S m^{hi}_{sj,t} dhdsdj$$

and

$$M^f_i = \int_0^S p_{sf,t} \int_0^S m^{hi}_{sf,t} dhds.$$

Then, our assumptions imply that the intermediate-good demand from sector hi for imported good s satisfies:

$$\frac{p_{sf,t}m^{hi}_{sf,t}}{M^f_i} = \frac{\int_0^I p_{sj,t}m^{hi}_{sj,t}dj}{M^d_i}.$$

Online Appendix

D Proof of Proposition 1

In equilibrium, N and M are linear in y as

$$N = (1-\alpha)(\lambda/w)y = (1-\alpha)\left(\frac{\sigma}{\sigma-1}\frac{P_m}{p}\right)^{\alpha/(1-\alpha)} y \tag{40}$$

and

$$M = \alpha(\lambda/P_m)y = \alpha\left(\frac{\sigma}{\sigma-1}\frac{P_m}{p}\right)^{-1} y. \tag{41}$$

The following equations determine the equilibrium.

$$\frac{p}{P} = \left(1+\xi_f\left(\frac{p}{p_f}\right)^{\sigma-1}\right)^{\frac{1}{\sigma-1}} =: \Gamma(p), \tag{42}$$

$$\frac{p}{P_m} = \left(1+\gamma_f\left(\frac{p}{p_f}\right)^{\sigma-1}\right)^{\frac{1}{\sigma-1}} =: \Gamma_m(p), \tag{43}$$

$$y = p^{-\sigma}\left(P^\sigma C + P_m^\sigma M + \omega^f\right), \tag{44}$$

$$PC = \mu_c p y, \tag{45}$$

$$\frac{w}{P} = \chi C^{\sigma_c} N^\zeta. \tag{46}$$

Then, the demand for domestic goods (44) is modified to

$$\left(1-\left(\mu_c\Gamma(p)^{1-\sigma}+(1-\mu_c)\Gamma_m(p)^{1-\sigma}\right)\right)p^\sigma y = \omega^f. \tag{47}$$

The left-hand side increases with p, so this equation represents a downward-sloping demand curve. More importantly, the demand curve shifts rightward with ω^f.

Note that (42) and (43) imply $p/P > 1$ and $p/P_m \geq 1$ (equality holds when $\gamma_f = 0$). Then, $1/(1-\mu_c\Gamma(p)^{1-\sigma}-(1-\mu_c)\Gamma_m(p)^{1-\sigma})$ acts as a "multiplier" on export sales $p^{1-\sigma}\omega^f$ to consumption expenditure PC, and it is larger when the domestic goods price p relative to the CPI P is lower.

If no imports are used as intermediate inputs, i.e., $\gamma_f = 0$, we have $\Gamma_m(p) = 1$ and (47) reduces to $\omega^f = (1-\Gamma(p)^{1-\sigma})\mu_c p^\sigma y$. If the use of imports in the intermediate sector is symmetric to that of domestic goods, i.e., $\gamma_f = 1$, we have $\Gamma_m(p) = \Gamma$ and (47) reduces to $\omega^f = (1-\Gamma(p)^{1-\sigma})p^\sigma y$.

The supply function for domestic goods is derived from (45) and (46). We use

$$\frac{w}{P} = \left(\frac{\sigma-1}{\sigma}\left(\frac{p}{P}\right)^{1-\alpha}\left(\frac{p}{P_m}\right)^{\alpha}\right)^{\frac{1}{1-\alpha}}.$$

This yields, with $y_o := ((\sigma-1)/\sigma)^{\frac{1+\alpha\zeta}{1-\alpha}}/(\chi(\mu_c)^{\sigma_c}(1-\alpha)^\zeta)$ as defined for (27),

$$(y_o)^{-1}y^{\sigma_c+\zeta} = \left(\frac{p}{P}\right)^{1-\sigma_c}\left(\frac{p}{P_m}\right)^{\frac{\alpha(1+\zeta)}{1-\alpha}} = \Gamma(p)^{1-\sigma_c}\Gamma_m(p)^{\frac{\alpha(1+\zeta)}{1-\alpha}}. \tag{48}$$

The supply function is upward-sloping if the wealth effect is not too strong, i.e., $\sigma_c < 1$. Even if the wealth effect is substantial ($\sigma_c > 1$), the supply function is upward-sloping if the effects through intermediate imports ($\gamma_f > 0$) are sufficiently strong.

If $\sigma_c \leq 1$, the right-hand side of (48) is finite and strictly increasing in p. In (47), as p increases from 0 to $+\infty$, y strictly decreases from $+\infty$ to 0. Hence, the solution p that satisfies both demand (47) and supply (48) exists and is unique. Moreover, since the right-hand side of (47) is strictly increasing in ω^f, we obtain $dp/d\omega^f > 0$ for $\sigma_c \leq 1$.

From (42), $d(p/P)/dp > 0$. Thus, $d(p/P)/d\omega^f \geq 0$ for $\sigma_c \leq 1$. Then, (48) implies $dy/d\omega^f \geq 0$ for $\sigma_c \leq 1$, with equality if $\sigma_c = 1$ and $\gamma_f = 0$. Moreover, since $C \propto (p/P)y$, we obtain $dC/d\omega^f > 0$.

Finally, as $\omega^f \searrow 0$, we obtain $p/P \to 1$ and $p \to 0$, completing the proof of item (a).

Item (b) is proved as follows. We set $\sigma_c = 1$. TS_c and TS_m denote the import shares of consumption and intermediate expenditure, respectively, and TS denotes the export share of total sales.

$$TS_c = 1 - \left(\frac{p}{P}\right)^{1-\sigma} \tag{49}$$

$$TS_m = 1 - \left(\frac{p}{P_m}\right)^{1-\sigma} \tag{50}$$

$$TS = \mu_c TS_c + (1-\mu_c) TS_m.$$

P and P_m are defined by (23, 24). The demand and supply curves (26,27) are given by

$$y = \frac{p^{-\sigma}\omega^f}{TS} \tag{51}$$

$$y = (y_o)^{\frac{1}{1+\zeta}} \left(\frac{p}{P_m}\right)^{\alpha/(1-\alpha)} \tag{52}$$

Thus, we have a system of equations that determines $(y, p, P, P_m, TS, TS_c, TS_m)$ for a given set of $(\sigma, \omega^f, \xi_f, \gamma_f, p_f, y_o, \alpha)$.

From (49,50), we obtain

$$\frac{d(TS_c)}{d(p/P)} = (\sigma - 1)\left(\frac{p}{P}\right)^{-\sigma} \tag{53}$$

$$\frac{d(TS_m)}{d(p/P_m)} = (\sigma - 1)\left(\frac{p}{P_m}\right)^{-\sigma} \tag{54}$$

Under the change of variable $x := p/P$, the definition of P in (23) yields $p^{1-\sigma}(x^{\sigma-1}-1) = \xi_f p_f^{1-\sigma}$. Differentiating yields $(1-\sigma)p^{-\sigma}(x^{\sigma-1}-1)dp + p^{1-\sigma}(\sigma-1)x^{\sigma-2}dx = 0$, which implies

$$\frac{dp}{d(p/P)} = \frac{px^{-1}}{1-x^{1-\sigma}} = \frac{P}{TS_c}. \tag{55}$$

Similarly, we obtain

$$\frac{dp}{d(p/P_m)} = \frac{P_m}{TS_m}. \tag{56}$$

From (53), (54), (55), and (56), we have

$$\frac{d(TS_c)}{d(TS_m)} = \frac{(p/P)^{-\sigma}}{(p/P_m)^{-\sigma}} \frac{d(p/P)}{d(p/P_m)} = \frac{(p/P)^{-\sigma}}{(p/P_m)^{-\sigma}} \frac{P_m/TS_m}{P/TS_c} = \left(\frac{P_m}{P}\right)^{1-\sigma} \frac{TS_c}{TS_m}. \tag{57}$$

Using the supply (52) and (56), we obtain

$$\begin{aligned}\frac{d(py)}{d(p/P_m)} &= (y_o)^{\frac{1}{1+\zeta}}\left[\frac{P_m}{TS_m}\left(\frac{p}{P_m}\right)^{\frac{\alpha}{1-\alpha}} + \frac{p\alpha}{1-\alpha}\left(\frac{p}{P_m}\right)^{\frac{\alpha}{1-\alpha}-1}\right] \\ &= (y_o)^{\frac{1}{1+\zeta}} P_m \left(\frac{p}{P_m}\right)^{\frac{\alpha}{1-\alpha}}\left(\frac{1}{TS_m} + \frac{\alpha}{1-\alpha}\right) \\ &= yP_m\left(\frac{1}{TS_m} + \frac{\alpha}{1-\alpha}\right). \end{aligned} \tag{58}$$

From (58) and (54), we obtain

$$\frac{d(TS_m)}{d(py)} = \frac{(\sigma-1)(p/P_m)^{-\sigma}}{yP_m\left(\frac{1}{TS_m}+\frac{\alpha}{1-\alpha}\right)} \tag{59}$$

We write the demand (51) as

$$d\log(p^{1-\sigma}\omega^f) = d\log(py) + d\log(TS).$$

Thus,

$$\frac{d\log(p^{1-\sigma}\omega^f)}{d\log(py)} = 1 + \frac{d\log(TS)}{d\log(py)}. \tag{60}$$

Hence, using (57) and (59), we obtain

$$\begin{aligned}\frac{d\log(TS)}{d\log(py)} &= \frac{py}{TS}\frac{(\mu_c d(TS_c)/d(TS_m) + 1 - \mu_c)d(TS_m)}{d(py)} \\ &= \frac{py}{TS}\left(\mu_c\left(\frac{P_m}{P}\right)^{1-\sigma}\frac{TS_c}{TS_m} + 1 - \mu_c\right)\frac{(\sigma-1)(p/P_m)^{-\sigma}}{yP_m\left(\frac{1}{TS_m}+\frac{\alpha}{1-\alpha}\right)} \\ &= \left(\mu_c\frac{1-TS_c}{1-TS_m}\frac{TS_c}{TS_m} + 1 - \mu_c\right)\frac{(\sigma-1)(1-TS_m)}{TS\left(\frac{1}{TS_m}+\frac{\alpha}{1-\alpha}\right)}. \end{aligned}$$

Substituting (60), we obtain the export-demand multiplier for aggregate sales as

$$\frac{d(py)}{d(p^{1-\sigma}\omega^f)} = \frac{d\log(py)}{d\log(p^{1-\sigma}\omega^f)}\frac{py}{p^{1-\sigma}\omega^f} = \frac{1}{TS + \left(\mu_c\frac{1-TS_c}{1-TS_m}\frac{TS_c}{TS_m} + 1 - \mu_c\right)\frac{(\sigma-1)(1-TS_m)}{\frac{1}{TS_m}+\frac{\alpha}{1-\alpha}}}$$

Hence, we obtain the desired result (25). □

E Analytical results on the first-order effect of an export shock

(i) Static model of Appendix F with $\sigma_c = 1$

Using the labor demand function, we have labor income $w_s^i N^{si} = (1-\alpha)\lambda^{si} y_{si} = ((1-\alpha)(\sigma-1)/\sigma)Y^{si}$, where we define the revenue of si as $Y^{si} := p_{si}y_{si}$. Letting regional total revenue $Y^i := \sum_s Y^{si}$, regional profits are $\Pi^i = (1-(\sigma-1)/\sigma)Y^i$. The international transfer associated with the trade deficit is B^i. We write its ratio to regional GDP as $b^i := B^i/Y^i$. From the household budget constraint, we have

$$P^i C^i = E^i = \sum_s (w_s^i N^{si}) + \Pi^i + B^i = (1-\alpha(\sigma-1)/\sigma + b^i)Y^i.$$

Aggregating the labor demand function, we have $\sum_s N^{si} w_s^i = ((1-\alpha)(\sigma-1)/\sigma)Y^i$. Using the labor supply function, we obtain

$$\frac{\sum_s n_s^i w_s^i}{P^i} = \chi_i (C^i)^{\sigma_c} (N^i)^{\zeta - 1/\tau} \sum_s (n_s^i)^{(1+\tau)/\tau} = \chi_i (C^i)^{\sigma_c} (N^i)^{\zeta+1}.$$

In equilibrium, $n_s^i = N^{si}$. Combining these equations yields

$$((1-\alpha)(\sigma-1)/\sigma)\frac{Y^i}{P^i} = \chi_i \left((1-\alpha(\sigma-1)/\sigma + b^i)\frac{Y^i}{P^i}\right)^{\sigma_c} (N^i)^{\zeta+1}.$$

Hence, when $\sigma_c = 1$, N^i is determined as $N^i = \bar{N}^i = \left(\frac{(1-\alpha)(\sigma-1)/\sigma}{\chi_i(1-\alpha(\sigma-1)/\sigma+b^i)}\right)^{\frac{1}{\zeta+1}}$, a constant independent of prices.

From the production function and the labor supply, we obtain

$$\begin{aligned} y_{si} &= A^{si}(M^{si})^{\alpha}(N^{si})^{1-\alpha} = A^{si}\left(\frac{\alpha}{1-\alpha}\frac{w_s^i}{P^{si}}\right)^{\alpha} N^{si} = A^{si}\left(\frac{\alpha}{1-\alpha}\frac{w_s^i}{P^{si}}\right)^{\alpha}\left(\frac{w_s^i}{P^i C^i \chi_i (\bar{N}^i)^{\zeta-1/\tau}}\right)^{\tau} \\ &= A^{si}\left(\frac{\alpha}{1-\alpha}\frac{w_s^i}{P^{si}}\right)^{\alpha}\left(\frac{w_s^i}{(1-\alpha(\sigma-1)/\sigma+b^i)\chi_i Y^i (\bar{N}^i)^{\zeta-1/\tau}}\right)^{\tau}. \end{aligned}$$

Also, w_s^i is a function of (P^{si}, p_{si}), since

$$\frac{\sigma-1}{\sigma} p_{si} = \lambda^{si} = \frac{(P^{si})^{\alpha}(w_s^i)^{1-\alpha}}{A^{si}\alpha^{\alpha}(1-\alpha)^{1-\alpha}}. \tag{61}$$

Multiplying p_{si} and y_{si}, we have

$$\begin{aligned} Y^{si} = p_{si}y_{si} &= \frac{\sigma/(\sigma-1)}{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}\left(\frac{\alpha}{1-\alpha}\right)^{\alpha}\frac{(w_s^i)^{1+\tau}}{\left((1-\alpha(\sigma-1)/\sigma+b^i)\chi_i Y^i(\bar{N}^i)^{\zeta-1/\tau}\right)^{\tau}} \\ &= \frac{(\hat{N}^i w_s^i)^{1+\tau}}{(Y^i)^{\tau}} \end{aligned} \tag{62}$$

where

$$\hat{N}^i := \left(\frac{\sigma/((\sigma-1)(1-\alpha))}{\left((1-\alpha(\sigma-1)/\sigma+b^i)\chi_i(\bar{N}^i)^{\zeta-1/\tau}\right)^{\tau}}\right)^{1/(1+\tau)} = \left(\frac{(\sigma/((\sigma-1)(1-\alpha)))^{\zeta}}{(1-\alpha(\sigma-1)/\sigma+b^i)\chi_i}\right)^{1/(1+\zeta)}.$$

Thus, we obtain Y^i as a function of $(w_s^i)_{s,i}$:

$$Y^i = \hat{N}^i \left(\sum_s (w_s^i)^{1+\tau} \right)^{1/(1+\tau)}. \tag{63}$$

The demand function for good sj implies

$$\begin{aligned} p_{sj}^{\sigma} y_{sj} &= \sum_i (P^i)^{\sigma} \xi_{sj}^i C^i + \sum_{h,i} (P^{hi})^{\sigma} \gamma_{sj}^{hi} M^{hi} + \omega_{sj}^f (\bar{P})^{\sigma} \\ &= \sum_i \left(1 - \alpha \frac{\sigma - 1}{\sigma} + b^i \right) (P^i)^{\sigma - 1} \xi_{sj}^i Y^i + \alpha \frac{\sigma - 1}{\sigma} \sum_{h,i} (P^{hi})^{\sigma - 1} \gamma_{sj}^{hi} p_{hi} y_{hi} + \omega_{sj}^f (\bar{P})^{\sigma}, \end{aligned} \tag{64}$$

using $P^i C^i = (1 - \alpha(\sigma - 1)/\sigma + b^i) Y^i$ and $P^{sj} M^{sj} = \alpha \lambda^{sj} y_{sj} = \alpha \frac{\sigma - 1}{\sigma} p_{sj} y_{sj}$.

Equations (61), (62), (63), and (64) determine $(w_s^i, Y^i, P^{si}, p_{si})_{s,i}$. (61) defines w_s^i as a function of (p_{si}, P^{si}). (63) determines Y^i as an aggregate of $(w_s^i)_s$. Substituting y_{si} from (62), the left-hand side of (64) becomes $p_{sj}^{\sigma - 1} Y^{sj}$, and the right-hand side depends on $(P^i, p_{hi})_{h,i}$. Moreover, the prices of consumption and intermediate composites (P^i, P^{hi}) satisfy

$$(P^i)^{1-\sigma} = \sum_{s,j} \xi_{sj}^i (p_{sj})^{1-\sigma} + S \xi_f^i (p_f)^{1-\sigma}$$

and

$$(P^{hi})^{1-\sigma} = \sum_{s,j} \gamma_{sj}^{hi} (p_{sj})^{1-\sigma} + \sum_s \gamma_{sf}^{hi} (p_{sf})^{1-\sigma}.$$

Thus, the modified (64) can be solved for an equilibrium price vector.

(ii) Comparative statics

We have a system of equations (64) that depends only on the vector $(p_{sj})_{s,j}$. This formulation allows us to conduct comparative statics with respect to $(p_{sj})_{s,j}$ when ω_{sj}^f is perturbed.

For ease of exposition, we denote the final goods sector by $s = 0$ and write $Y^{0i} := Y^i$. We introduce a new weight matrix $\hat{\Gamma}$, whose elements are defined as

$$\hat{\gamma}_{sj}^{0i} = \left(1 - \alpha \frac{\sigma - 1}{\sigma} + b^i \right) \xi_{sj}^i$$

and

$$\hat{\gamma}_{sj}^{hi} = \alpha \frac{\sigma - 1}{\sigma} \gamma_{sj}^{hi}.$$

Using (61), (62), (63), and (64), we have the following system for $(p_{si}, P^{si}, Y^{si}, w_s^i)_{si}$:

$$p_{si}^{\sigma - 1} Y^{si} = \sum_{h=0}^{S} \sum_{j=1}^{I} \hat{\gamma}_{si}^{hj} (P^{hj})^{\sigma - 1} Y^{hj} + \omega_{si}^f (\bar{P})^{\sigma}. \tag{65}$$

$$Y^{si} = \hat{N}^i \frac{(w_s^i)^{1+\tau}}{\left(\sum_{h=1}^{S} (w_h^i)^{1+\tau} \right)^{\tau/(1+\tau)}} \quad \text{for } s = 1, 2, \ldots, S, \tag{66}$$

$$Y^{0i} = \hat{N}^i \left(\sum_{h=1}^{S} (w_h^i)^{1+\tau} \right)^{1/(1+\tau)}, \tag{67}$$

$$(P^{si})^{1-\sigma} = \sum_{h=1}^{S}\sum_{j=1}^{I}\gamma_{hj}^{si}(p_{hj})^{1-\sigma} + \sum_{h=1}^{S}\gamma_{hf}^{si}(p_{hf})^{1-\sigma} \quad \text{for } s = 1, 2, \ldots, S, \tag{68}$$

$$(P^{0i})^{1-\sigma} = \sum_{h=1}^{S}\sum_{j=1}^{I}\xi_{hj}^{i}(p_{hj})^{1-\sigma} + S\xi_{f}^{i}(p_{f})^{1-\sigma}, \tag{69}$$

and

$$w_s^i = \left(\frac{\sigma-1}{\sigma}\alpha^{\alpha}(1-\alpha)^{1-\alpha}A^{si}p_{si}(P^{si})^{-\alpha}\right)^{1/(1-\alpha)} \quad \text{for } s = 1, 2, \ldots, S. \tag{70}$$

Consumption is determined by $C^i = (1-\alpha(\sigma-1)/\sigma + b^i)Y^{0i}/P^{0i}$.

These equations allow for the analytical expression of the elasticity $d\ln p_{hj}/d\ln\omega_{si}^f$. To simplify, we change variables to $q_{si} := p_{si}^{\sigma-1}$ and $Q^{si} := (P^{si})^{\sigma-1}$. From (68), we have $d\ln Q^{si}/d\ln q_{hj} = Q^{si}\gamma_{hj}^{si}/q_{hj}$. With the new notation (q_{si}, Q^{si}), (65) is written as $q_{si}Y^{si} = \sum_{h=0}^{S}\sum_{j=1}^{I}\hat{\gamma}_{si}^{hj}Q^{hj}Y^{hj} + \omega_{si}^f(\bar{P})^{\sigma}$. By log-differentiation, we obtain:

$$q_{si}Y^{si}(d\ln q_{si} + d\ln Y^{si}) = \sum_{h=1}^{S}\sum_{j=1}^{I}\hat{\gamma}_{si}^{hj}Q^{hj}Y^{hj}(d\ln Q^{hj} + d\ln Y^{hj}) + dD_{si}^{c} + \omega_{si}^{f}(\bar{P})^{\sigma}d\ln\omega_{si}^{f}, \tag{71}$$

where $D_{si}^{c} := \sum_{j=1}^{I}\hat{\gamma}_{si}^{0j}Q^{0j}Y^{0j}$ corresponds to consumption demand.

Note that (66), (70), and the definition of Q imply:

$$d\ln Y^{si} = (1+\tau)d\ln w_s^i - \tau\sum_{s'}\frac{(w_{s'}^i)^{1+\tau}d\ln w_{s'}^i}{\sum_{s''}(w_{s''}^i)^{1+\tau}} + d\ln\hat{N}^i,$$

$$d\ln\hat{N}^i = \frac{-db^i}{(1+\zeta)(1-\alpha(\sigma-1)/\sigma + b^i)}$$

$$d\ln w_s^i = \frac{d\ln p_{si} - \alpha d\ln P^{si}}{1-\alpha} = \frac{d\ln q_{si} - \alpha\sum_{s',i'}\frac{\gamma_{s'i'}^{si}Q^{si}}{q_{s'i'}}d\ln q_{s'i'}}{(\sigma-1)(1-\alpha)},$$

and

$$d\ln Q^{si} = Q^{si}\sum_{s',i'}(\gamma_{s'i'}^{si}/q_{s'i'})d\ln q_{s'i'}.$$

Vectorize the $S\times I$ matrices into $SI\times 1$ vectors and denote them by $\boldsymbol{q}$, $\boldsymbol{w}$, $\boldsymbol{Y}$, $\boldsymbol{Q}$, $\boldsymbol{\omega}^f$, and $\boldsymbol{D^c}$. Also vectorize $\mathbf{1}_{S\times 1}\cdot(\hat{N}^i)_i$ and $\mathbf{1}_{S\times 1}\cdot(b^i)_i$ and denote them by $\hat{\boldsymbol{N}}$ and $\boldsymbol{b}$, (i.e., each $\hat{N}^i$ (b^i) is duplicated S times and stacked into a column). Let $\mathrm{diag}\,(qY)$ denote a diagonal matrix with $q_{si}Y^{si}$ at position $(S(i-1)+s, S(i-1)+s)$. Similarly, $\mathrm{diag}\,(Q)$ and $\mathrm{diag}\,(\omega^f)$ denote diagonal matrices with Q^{si} and ω_{si}^f, respectively.

Let W_I denote an $I\times SI$ matrix with entry $(i, S(i-1)+s)$ equal to $(w_s^i)^{1+\tau}/\sum_{s'}(w_{s'}^i)^{1+\tau}$. Let W denote an $SI\times SI$ matrix in which the row vector $W_I(i, S(i-1)+s)$, for fixed i, is duplicated across the rows from $S(i-1)+1$ to Si. Moreover, set G_w as a $SI\times SI$ matrix with $\gamma_{s'i'}^{si}/q_{s'i'}$ in position $(S(i-1)+s, S(i'-1)+s')$.

The equations above are written as follows.

$$d\ln\boldsymbol{Y} = ((1+\tau)I_{SI} - \tau W)d\ln\boldsymbol{w} + d\ln\hat{\boldsymbol{N}},$$

$$d\ln\boldsymbol{w} = \frac{(I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w)d\ln\boldsymbol{q}}{(\sigma-1)(1-\alpha)},$$

and

$$d\ln \boldsymbol{Q} = \mathrm{diag}\,(Q) G_w d\ln \boldsymbol{q},$$

where I_{SI} denotes the $SI \times SI$ identity matrix. In short,

$$d\ln \boldsymbol{Y} = \frac{((1+\tau)I_{SI} - \tau W)(I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w)}{(\sigma-1)(1-\alpha)} d\ln \boldsymbol{q} + d\ln \hat{\boldsymbol{N}}.$$

Furthermore, define G as an $SI \times SI$ matrix with $\hat{\gamma}_{si}^{s'i'} Q^{s'i'} Y^{s'i'}$ in position $(S(i-1)+s, S(i'-1)+s')$. Then, (71) can be written as

$$\mathrm{diag}\,(qY)(d\ln \boldsymbol{q} + d\ln \boldsymbol{Y}) = G(d\ln \boldsymbol{Q} + d\ln \boldsymbol{Y}) + d\boldsymbol{D^c} + \mathrm{diag}\,(\omega^f)(\bar{P})^\sigma d\ln \boldsymbol{\omega^f}. \tag{72}$$

This equation yields a decomposition.

$$d\ln \boldsymbol{Y} = \mathrm{diag}\,(qY)^{-1}\left[Gd\ln \boldsymbol{Y} + (Gd\ln \boldsymbol{Q} - \mathrm{diag}\,(qY)d\ln \boldsymbol{q}) + d\boldsymbol{D^c} + \mathrm{diag}\,(\omega^f)(\bar{P})^\sigma d\ln \boldsymbol{\omega^f}\right]. \tag{73}$$

Response of consumption The consumption term is $D^c_{si} = \sum_{j=1}^{I} \hat{\gamma}_{si}^{0j} Q^{0j} Y^{0j}$. Let $\boldsymbol{Y}^0$ be the vector $(Y^{0i})_i$, $\boldsymbol{Q}^0$ the vector $(Q^{0i})_i$, and Γ^0 an $SI \times I$ matrix with entries $\hat{\gamma}_{si}^{0j} Q^{0j} Y^{0j}$ at position $(S(i-1)+s, j)$. Note that $\hat{\gamma}_{sj}^{0i}$ includes b^i. Let $\boldsymbol{b}_I$ denote the vector $(b^1, b^2, \ldots, b^I)'$. Also, let $\hat{\boldsymbol{N}}_I$ denote the vector $(\hat{N}^1, \hat{N}^2, \ldots, \hat{N}^I)'$. Then, we can write

$$d\boldsymbol{D^c} = \Gamma^0 \left(d\ln \boldsymbol{Q}^0 + d\ln \boldsymbol{Y}^0 + \mathrm{diag}\left(\frac{1}{1-\alpha(\sigma-1)/\sigma + b_I}\right) d\boldsymbol{b}_I \right)$$

Plugging into (73), we obtain a decomposition equation:

$$\begin{aligned} d\ln \boldsymbol{Y} = \mathrm{diag}\,(qY)^{-1} \big[& Gd\ln \boldsymbol{Y} + (Gd\ln \boldsymbol{Q} - \mathrm{diag}\,(qY)d\ln \boldsymbol{q} + \Gamma^0 d\ln \boldsymbol{Q}^0) \\ & + \Gamma^0 d\ln \boldsymbol{Y}^0 + \Gamma^0\,\mathrm{diag}\left(\frac{1}{1-\alpha(\sigma-1)/\sigma + b_I}\right) d\boldsymbol{b}_I + \mathrm{diag}\,(\omega^f)(\bar{P})^\sigma d\ln \boldsymbol{\omega^f} \big] \end{aligned} \tag{74}$$

We further proceed with

$$\begin{aligned} Y^{0i} &= \hat{N}^i \left(\sum_{s=1}^{S} (w_s^i)^{1+\tau} \right)^{1/(1+\tau)}, \\ (Q^{0j})^{-1} &= \sum_{s=1}^{S} \sum_{i=1}^{I} \xi_{si}^j q_{si}^{-1} + S\xi_f^j (p_f)^{1-\sigma}. \end{aligned}$$

From these, we have

$$\begin{aligned} d\ln Y^{0i} =& \frac{\sum_s (w_s^i)^{1+\tau} d\ln w_s^i}{\sum_{s'} (w_{s'}^i)^{1+\tau}} + d\ln \hat{N}^i \\ d\ln Q^{0j} =& Q^{0j} \sum_{s,i} \left(\xi_{si}^j q_{si}^{-1} d\ln q_{si} \right). \end{aligned}$$

Then,

$$d\ln \boldsymbol{Y}^0 = W_I d\ln \boldsymbol{w} + d\ln \hat{N}_I = W_I \frac{I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w}{(\sigma-1)(1-\alpha)} d\ln \boldsymbol{q} + d\ln \hat{N}_I.$$

Also, let matrix $\Xi_{I\times SI}$ contain $Q^{0j}\xi^j_{si}q^{-1}_{si}$ in $(j, S(i-1)+s)$. Then we obtain

$$d\ln \boldsymbol{Q}^0 + d\ln \boldsymbol{Y}^0 = \left(\frac{W_I(I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w)}{(\sigma-1)(1-\alpha)} + \Xi\right) d\ln \boldsymbol{q} + d\ln \hat{N}_I.$$

Using (72),

$$\begin{aligned}
&\mathrm{diag}\,(qY)\left(I_{SI} + \frac{((1+\tau)I_{IS} - \tau W)(I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w)}{(\sigma-1)(1-\alpha)}\right) d\ln \boldsymbol{q} + \mathrm{diag}\,(qY) d\ln \hat{\boldsymbol{N}} \\
&= G\left(\mathrm{diag}\,(Q)G_w + \frac{((1+\tau)I_{IS} - \tau W)(I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w)}{(\sigma-1)(1-\alpha)}\right) d\ln \boldsymbol{q} + G d\ln \hat{\boldsymbol{N}} \\
&+ d\boldsymbol{D^c} + \mathrm{diag}\,(\omega^f)(\bar{P})^\sigma d\ln \boldsymbol{\omega^f}
\end{aligned}$$

or,

$$\begin{aligned}
d\ln \boldsymbol{q} = &\left[\mathrm{diag}\,(qY) - G\,\mathrm{diag}\,(Q)G_w + (\mathrm{diag}\,(qY) - G)\,\frac{((1+\tau)I_{IS} - \tau W)(I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w)}{(\sigma-1)(1-\alpha)}\right]^{-1} \\
&\cdot \left(d\boldsymbol{D^c} + \mathrm{diag}\,(\omega^f)(\bar{P})^\sigma d\ln \boldsymbol{\omega^f} + (G - \mathrm{diag}\,(qY)) d\ln \hat{\boldsymbol{N}}\right). \qquad (75)
\end{aligned}$$

Construct an $SI \times SI$ matrix G^0 such that its $(S(i-1)+1)$-th column equals the i-th column of Γ^0, and all other columns are zero. Substituting out $d\boldsymbol{D^c}$ in (75) yields

$$\begin{aligned}
d\ln \boldsymbol{q} = &\Bigg[\mathrm{diag}\,(qY) - G\,\mathrm{diag}\,(Q)G_w \\
&+ (\mathrm{diag}\,(qY) - G)\,\frac{((1+\tau)I_{IS} - \tau W)(I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w)}{(\sigma-1)(1-\alpha)} \\
&- \Gamma^0\left(\frac{W_I(I_{SI} - \alpha\,\mathrm{diag}\,(Q)G_w)}{(\sigma-1)(1-\alpha)} + \Xi\right)\Bigg]^{-1} \\
&\cdot \left[\mathrm{diag}\,(\omega^f)(\bar{P})^\sigma d\ln \boldsymbol{\omega^f} + \left(\frac{G - \mathrm{diag}\,(qY) - \zeta G^0}{1+\zeta}\right)\mathrm{diag}\left(\frac{-1}{1-\alpha(\sigma-1)/\sigma + b}\right) d\boldsymbol{b}\right]. \qquad (76)
\end{aligned}$$

The real output is $\bar{Y}^i := \sum_s \bar{p}_{si} y_{si} = \sum_s Y_{si}(\bar{q}_{si}/q_{si})^{1/(\sigma-1)}$. Hence, evaluated at the reference equilibrium, $d\ln \bar{Y}^i = \sum_s (Y_{si}/\bar{Y}^i)(d\ln Y_{si} - (1/(\sigma-1)) d\ln q_{si})$.

F Static model

The baseline model presented in the main text (Section 3) is a dynamic model that incorporates investment. In this section, we consider a static model to contrast with the results in the main text. Although some portions are straightforward modifications of the baseline model, we allow some overlap with the main text to provide a self-contained exposition of the static model.

(i) Model setting

The setting is identical to the baseline model, except that capital stock and investment are absent. Consider a small open economy with I regions. In each region, there are S industries. Thus, an industry is indexed by (s,i), where $s \in [0,S]$ and $i \in [0,I]$. Each industry (s,i) is monopolistically competitive; that is, only one firm produces in industry (s,i). Producing a good requires labor and intermediate goods as inputs. Product (s,i) is used for consumption, as an intermediate good in production, and for export. Each region i has a representative consumer who owns the firm in region i, supplies labor to the firms in region i, and consumes both domestic and imported goods.

(A) Representative consumer

The representative consumer in region i maximizes utility

$$U^i = \frac{(C^i)^{1-\sigma_c}-1}{1-\sigma_c} - \chi_i \frac{(N^i)^{1+\zeta}}{1+\zeta}$$

subject to

$$P^i C^i \leq E^i := \int_0^S w_s^i n_s^i ds + \Pi^i + B^i, \tag{77}$$

where

$$C^i = \left[\int_0^S \int_0^I (\xi_{sj}^i)^{\frac{1}{\sigma}} (c_{sj}^i)^{\frac{\sigma-1}{\sigma}} dj ds + \int_0^S (\xi_{sf}^i)^{\frac{1}{\sigma}} (c_{sf}^i)^{\frac{\sigma-1}{\sigma}} ds\right]^{\frac{\sigma}{\sigma-1}}$$

and

$$N^i = \left[\int_0^S (n_s^i)^{\frac{\tau+1}{\tau}} ds\right]^{\frac{\tau}{\tau+1}}.$$

The notation is the same as in our baseline formulation. The variable E^i denotes the expenditure of consumer i.

The consumer's optimization yields the labor supply relationship

$$\frac{w_s^i}{P^i} = \chi_i (C^i)^{\sigma_c} (N^i)^{\zeta} \left(\frac{n_s^i}{N^i}\right)^{\frac{1}{\tau}}. \tag{78}$$

To consume the goods, the consumer allocates consumption across goods by solving the expenditure-minimization problem

$$\min_{c_{sj}^i, c_{sf}^i} \quad \int_0^S \int_0^I p_{sj} c_{sj}^i d_j ds + \int_0^S p_f c_{sf}^i ds$$

subject to

$$C^i = \left[\int_0^S \int_0^I (\xi_{sj}^i)^{\frac{1}{\sigma}} (c_{sj}^i)^{\frac{\sigma-1}{\sigma}} dj ds + \int_0^S (\xi_f^i)^{\frac{1}{\sigma}} (c_{sf}^i)^{\frac{\sigma-1}{\sigma}} ds\right]^{\frac{\sigma}{\sigma-1}}.$$

Here, p_{sj} is the price of good (s,j), which is the same across regions. The prices of imported goods are assumed to be uniform at p_f. The solution to the optimization yields the demand for domestic goods

$$c_{sj}^i = \left(\frac{p_{sj}}{P^i}\right)^{-\sigma} \xi_{sj}^i C^i,$$

and for imported foreign goods

$$c_{sf}^i = c_f^i = \left(\frac{p_f}{P^i}\right)^{-\sigma} \xi_f^i C^i, \tag{79}$$

where the price index is given by

$$P^i := \left[\int_0^S \int_0^I \xi_{sj}^i (p_{sj})^{1-\sigma} djds + S\xi_f^i (p_f)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}. \tag{80}$$

(B) Production

In region i, good h is produced according to the production function

$$y_{hi} = A^{hi}(M^{hi})^{\alpha}(N^{hi})^{1-\alpha},$$

where

$$M^{hi} = \left[\int_0^S \int_0^I (\gamma_{sj}^{hi})^{\frac{1}{\sigma}} (m_{sj}^{hi})^{\frac{\sigma-1}{\sigma}} djds + \int_0^S (\gamma_{sf}^{hi})^{\frac{1}{\sigma}} (m_{sf}^{hi})^{\frac{\sigma-1}{\sigma}} ds\right]^{\frac{\sigma}{\sigma-1}}.$$

Here, m_{sj}^{hi} is the intermediate good s from region j used in producing good h in region i, and γ_{sj}^{hi} is a parameter. Similarly, m_{sf}^{hi} is the imported intermediate good s used in producing good h in region i, and γ_{sf}^{hi} is a parameter.

The demand function for intermediate goods is

$$m_{sj}^{hi} = \left(\frac{p_{sj}}{P^{hi}}\right)^{-\sigma} \gamma_{sj}^{hi} M^{hi} \quad \text{for } j \in [0,I] \cup \{f\},$$

where

$$P^{hi} := \left[\int_0^S \int_0^I \gamma_{sj}^{hi} (p_{sj})^{1-\sigma} djds + \int_0^S \gamma_{sf}^{hi} (p_{sf})^{1-\sigma} ds\right]^{\frac{1}{1-\sigma}}. \tag{81}$$

Thus, the total demand for good (s,j), obtained by adding consumption and intermediate-good demand, is

$$\begin{aligned} y_{sj} &= \int_0^I c_{sj}^i di + \int_0^S \int_0^I m_{sj}^{hi} didh + y_{sj}^f \\ &= (p_{sj})^{-\sigma} \left(\int_0^I (P^i)^{\sigma} \xi_{sj}^i C^i di + \int_0^S \int_0^I (P^{hi})^{\sigma} \gamma_{sj}^{hi} M^{hi} didh\right) + y_{sj}^f, \end{aligned}$$

where y_{sj}^f represents foreign (export) demand. Assume that foreign demand takes the form

$$y_{sj}^f = \omega_{sj}^f (p_{sj})^{-\sigma} (\bar{P})^{\sigma}, \tag{82}$$

that is, foreign demand has the same price elasticity as domestic demand. $\bar{P}$ is the price level in the foreign country.

Let

$$D_{sj} := \left(\int_0^I (P^i)^{\sigma} \xi_{sj}^i C^i di + \int_0^S \int_0^I (P^{hi})^{\sigma} \gamma_{sj}^{hi} M^{hi} didh + \omega_{sj}^f (\bar{P})^{\sigma}\right) \tag{83}$$

so that the demand for good (s,j) is $(p_{sj})^{-\sigma}D_{sj}$. We analyze the firm's problem in two steps. First, the firm chooses the input combination to minimize unit cost:

$$\min_{M^{sj},N^{sj}} \; P^{sj}M^{sj} + w_s^j N^{sj}$$

subject to

$$1 = A^{sj}(M^{sj})^{\alpha}(N^{sj})^{1-\alpha}.$$

The solution yields the unit cost λ^{sj}:

$$\lambda^{sj} = \frac{(P^{sj})^{\alpha}(w_s^j)^{1-\alpha}}{A^{sj}\alpha^{\alpha}(1-\alpha)^{1-\alpha}} \tag{84}$$

and the derived factor demands for unit output:

$$M^{sj,1} = \frac{\alpha}{P^{sj}}\lambda^{sj},$$

$$N^{sj,1} = \frac{1-\alpha}{w_s^j}\lambda^{sj}.$$

Second, the firm maximizes profit:

$$\max_{p_{sj}} \; (p_{sj} - \lambda^{sj})(p_{sj})^{-\sigma}D_{sj}.$$

The result is the standard constant markup rule:

$$p_{sj} = \frac{\sigma}{\sigma-1}\lambda^{sj}. \tag{85}$$

Thus, the production of good (s,j) is

$$y_{sj} = \left(\frac{\sigma}{\sigma-1}\lambda^{sj}\right)^{-\sigma} D_{sj}. \tag{86}$$

The derived factor demand can therefore be computed as:

$$M^{sj} = \frac{\alpha}{P^{sj}}\lambda^{sj}y_{sj} \tag{87}$$

and

$$N^{sj} = \frac{1-\alpha}{w_s^j}\lambda^{sj}y_{sj}. \tag{88}$$

(C) Trade balance

As in the baseline model, we allow for a trade imbalance. At the national level, the international account is

$$\int_0^I B^i di = \int_0^S \int_0^I p_{sf}\left(c_{sf}^i + \int_0^S m_{sf}^{hi} dh\right) dids - \int_0^S \int_0^I p_{si}y_{si}^f dids.$$

(D) Equilibrium

The labor market equilibrium requires

$$N^{si} = n_s^i$$

for all (s,i). The total profit is

$$\Pi^i = \int_0^S (p_{si} - \lambda^{si})y_{si}ds. \tag{89}$$

Using these two conditions and the budget constraint, we can compute the equilibrium value of C^i.

(ii) Computation and calibration

(A) Computation

The equilibrium of the model economy is computed using the following steps. Note that this model is static, so we can compute it period by period.

1. Normalize the foreign price level $\bar{P} = 1$ and the import price $p_f = 1$.

2. Guess w_s^i for all (s, i) and p_{si} for all (s, i). Then compute price indices P^i and P^{si} from (80) and (81). Compute the unit cost λ^{si} from (84), then verify p_{si} using the markup formula (85). Thus, for a given (w_s^i), this routine yields p_{si} consistent with it.

3. Further guess D_{si} for all (s, i). Then compute y_{sj}, M^{sj}, N^{sj} using (86), (87), (88). Compute Π^i from (89). Then compute $E^i = \sum_{s=1}^{S} w_s^i n_s^i + \Pi^i + B^i$. Use the budget constraint (77) to compute C^i, and use M^{sj} and C^i in (83) to check whether the initial guess on D_{si} was correct.

4. Finally, check (w_s^i) using (78).

(B) Calibration

Calibration is similar to the baseline model. We start from the economy in 2008Q3, just before the export shock hits. The consumption share parameters $\{\xi_{sj}^i\}_{i,sj}$ are calibrated so that the consumption expenditure share of good (s, j), representing good s produced in region j and consumed by consumers in region i, matches the data in the baseline economy. The consumption shares are taken from the inter-regional input-output table for 2005 (IRIO2005), the closest time period before 2008Q3. Similarly, we set the target for $\{\xi_i^f\}_i$ as the GDP share of export goods produced in region i. Parameters governing foreign demand for the (s, j) production, $\{\omega_{sj}^f\}_{sj}$, are set so that the GDP share of export goods (s, j) matches the data computed in IRIO2005.[17]

We assume that the parameter governing the wage elasticity of labor supply choice in each industry, τ, equals 1, as in Horvath (2000). The inverse of the Frisch elasticity of overall labor supply, ζ, is set to 2.5, based on Kuroda and Yamamoto (2008).[18] The labor disutility parameter χ_i is calibrated to replicate the regional variation in the employed population in 2008Q3, drawn from the Labour Force Survey (LFS) conducted by the Statistics Bureau of Japan.[19] Note that variation in the employed population reflects variation in the labor force (or working-age population) and the employment rate.[20] As a benchmark, we consider the case of $\sigma_c \to 1$.

The elasticity of substitution parameter, σ, is set to 5. The parameters governing the cost share of each intermediate good (s, j) for the producer of good h in region i, $\{\gamma_{sj}^{hi}\}_{hi,sj}$, are set to match the benchmark values to the corresponding data in IRIO2005. Factor-neutral productivity for each industry (s, j), A_{sj}, is given by the product of the industry- and region-specific productivity parameters; that is, $A_{sj} = A_s \times A_j$, where A_s is the industry-specific productivity and A_j is the

[17] As a result, the export-to-GDP ratio and the share of good (s, j) in total exports match the data.

[18] Kuroda and Yamamoto (2008) estimate the Frisch elasticity in Japan and report that the elasticity on the extensive and intensive margins combined ranges between 0.2 and 0.7 for males. $\zeta = 2.5$ implies a Frisch elasticity of 0.4.

[19] See https://www.stat.go.jp/english/index.html.

[20] Although it is better to incorporate variation in working hours per labor force, there are no reliable data disaggregating working hours by region.

Parameter	Description	Value	Target/Source
Preference			
σ_c	curvature	1.0	Assumed
χ_i	disutility of labor supply	Table 5	LFS (2008)
ζ	inverse of Frisch elasticity	2.5	Kuroda and Yamamoto (2008)
τ	elasticity of substitution (labor)	1.0	Benchmark in Horvath (2000)
$\{\xi^i_{sj}\}_{i,sj}$	weight on consumption goods	Figure 18a	IRIO (2005)
$\{\omega^f_{sj}\}_{sj}$	weight on export goods	Figure 18c	IRIO (2005)
Technology			
$\{A_{sj}\}_{sj}$	factor neutral productivity	Tables 4 and 5	JIP (2008), MLS (2008)
$\{\alpha_s\}_s$	cost share of intermediate goods	Table 4	JIP (2008)
σ	elasticity of substitution	5.0	Assumed
$\{\gamma^{hi}_{sj}\}_{hi,sj}$	weight on intermediate goods	Figure 18b	IRIO (2005)

Table 8: Summary of the parameter values, their source/reference, and data for setting targets.

region-specific productivity. First, we map the industry classification in the JIP Database to ours and compute the industrial TFP and the cost share of intermediate goods for each industry s ($\{\alpha_s\}_s$). Given all other parameters, the region-specific productivity A_j is pinned down so that the regional variation in the average wage rate in the benchmark replicates the data counterpart computed using the Monthly Labour Survey (MLS).

Table 8 summarizes the parameter values. The values of A_s and $\{\alpha_s\}_s$ are shown in Table 4 in Appendix C. The regional parameters A_i and χ_i are shown in Table 5 in Appendix C. The parameters $\{\xi^i_{sj}\}_{i,sj}$, $\{\gamma^{hi}_{sj}\}_{hi,sj}$, and $\{\omega^f_{sj}\}_{sj}$ are too numerous to summarize in a table and are shown as heatmaps in Figure 18 in Appendix C.

(iii) Simulating the model with the Great Recession export shocks

We repeat the experiments from the main text. Below, we set the time series for y^f_{sj} so that the time path of the export value matches the regional export data.

We simulate export shocks to industry si (i.e., industry s in region i) in period t by adjusting $y^f_{si,t}$ so that the following holds in equilibrium:

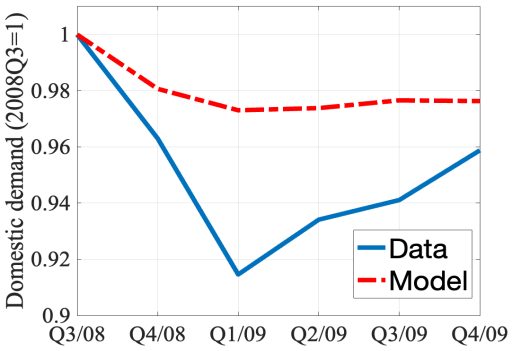
$$\frac{y^f_{si,t}}{y^f_{si,t=0}} = \frac{\text{real export of } si \text{ in } t \text{ in data}}{\text{real export of } si \text{ in } t=0 \text{ in data}}, \tag{90}$$

where the reference period $t = 0$ corresponds to the third quarter of 2008 (2008Q3).

Our primary interest is domestic (final) demand, equivalent to consumption in this static model, at the national and regional levels. Real consumption is computed excluding the consumption of imported goods. Formally, the real consumption for region i in period t, $\bar{C}_{i,t}$, is defined as

$$\bar{C}_{i,t} = \int_0^S \int_0^I p_{sj,t=0} c^i_{sj,t} dj ds, \tag{91}$$

(a) Domestic final demand



(b) Exports

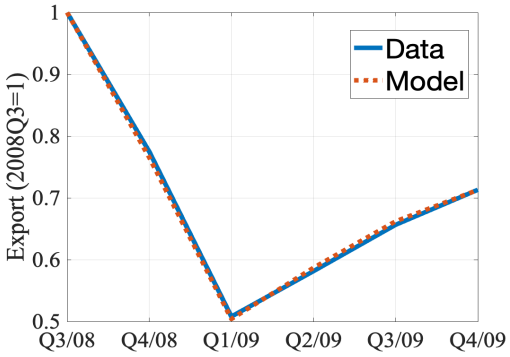


Figure 20: National responses. "Data" plots the fluctuations of HP-filtered variables, normalizing 2008Q3's value as 1.

where $p_{sj,t=0}$ is the price of goods produced in industry sj in the reference period $t = 0$. The national (real) consumption in period t, $\bar{C}_{Japan,t}$, is then defined as

$$\bar{C}_{Japan,t} = \int_0^I \bar{C}_{i,t} di. \tag{92}$$

(A) National level response

Figure 20 shows domestic final demand and exports at the national level. By construction, the model's export values exactly match the data. The model accounts for 63.2% of the decline in consumption in 2009Q1 and 19.5% of the decline in average consumption from 2008Q4 to 2009Q4. The static model's demand decline is more modest than that of the dynamic model because it does not capture investment. According to the data, investment experienced a substantial decline, which contributed significantly to the overall decline in domestic demand and GDP.

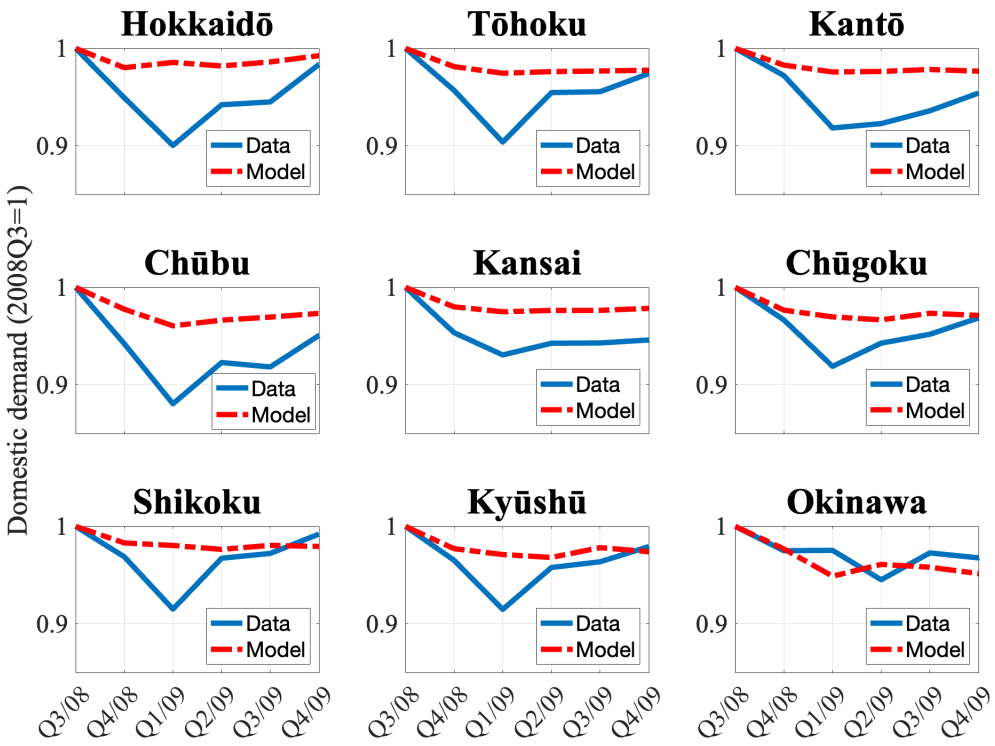


Figure 21: Regional responses

(B) Responses at the regional level

Figure 21 compares the model's regional demand for domestic final goods with the data. The model fits the data particularly well for regions such as Chūbu, Kantō, and Kansai, which experienced large declines in exports. In contrast, for regions such as Shikoku and Okinawa, where export shocks were not significant, the model performs poorly at explaining the regional decline in GDP.

(iv) Counterfactual experiments

In this section, we conduct a controlled experiment by feeding the model only the shock to y^f_{sj} for one region and industry, while holding y^f_{sj} for the other regions and industries constant. As in the main text, we consider a negative export shock to the transportation equipment (TE) industry in Chūbu.

(A) Decomposition

To assess how an export-demand shock in a region affects other regions, we conduct a decomposition analysis. Our decomposition is based on distinct demand components. In the following equation, the first term on the right-hand side represents domestic consumption demand, the second term represents domestic intermediate-good demand, and the third term represents foreign demand for goods s produced in region j.

$$y_{sj} = \int_0^I c^i_{sj} di + \int_0^S \int_0^I m^{hi}_{sj} didh + y^f_{sj} \tag{93}$$

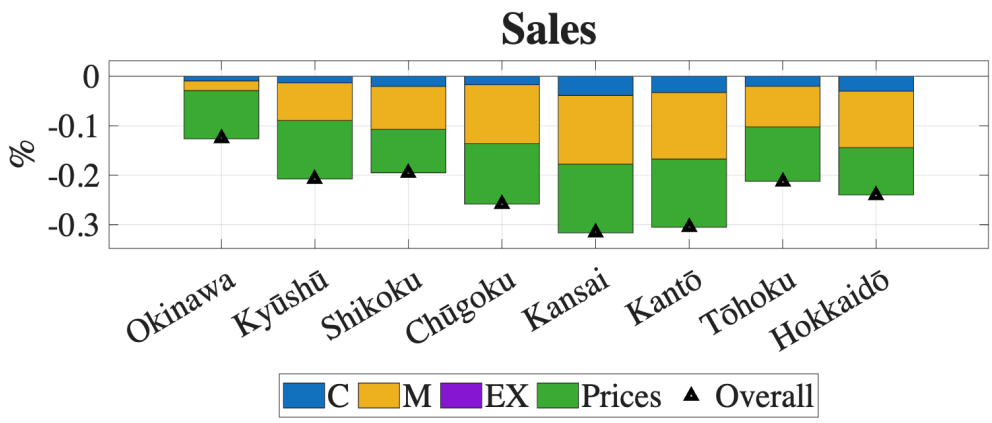


Figure 22: Changes in output with 2009Q1's shock to the TE in Chūbu (2008Q3=1).

Domestic consumption demand is given by

$$c_{sj}^{i} = \xi_{sj}^{i} \left(\frac{p_{sj}}{P^{i}} \right)^{-\sigma} C^{i}, \tag{94}$$

and domestic intermediate-good demand from industry h in region i is

$$m_{sj}^{hi} = \gamma_{sj}^{hi} \left(\frac{p_{sj}}{P^{hi}} \right)^{-\sigma} M^{hi}. \tag{95}$$

Given this demand structure, we compare two economies. The first is the baseline economy (2008Q3) with no shocks. The second is the economy that experienced an export shock in 2009Q1, affecting only one industry and one region. As mentioned above, we chose the transportation equipment industry in the Chūbu region. Comparing these two economies shows the overall changes in (real) output in each region under this particular shock, where the real output for region i is formulated as follows:

$$\bar{Y}_i = \int_0^S p_{si,t=0} y_{si} ds. \tag{96}$$

Our decomposition exercise involves two steps. In the first step, we decompose the output change in each region into several demand factors. More specifically, we consider the following three factors separately when decomposing the output change in region i. That is, we change only one of these factors in equations (94) and (95). The first set of factors represents the effect of *prices*. These are p_{sj} for all s and j in (94) and (95), and the price indices P^i and P^{hi} in (94) and (95). Note that the foreign price $\bar{P}$ is fixed because of the small open economy assumption. The second and third factors are C^i and M^{hi}. The first step reveals which factor affects a region's output but is silent about which region. The second step then decomposes the contribution of each factor into the regions from which those effects originate.

Figure 22 shows the decomposition result for the first step across regions other than Chūbu. The overall effect is negative in every region, as indicated by triangles. The closer a region is to Chūbu, the more pronounced its overall decline tends to be.

Figures 23 and 24 show, respectively, the contributions of each region to the column region's declines in consumption and intermediate goods demand. Across regions, the decline in each

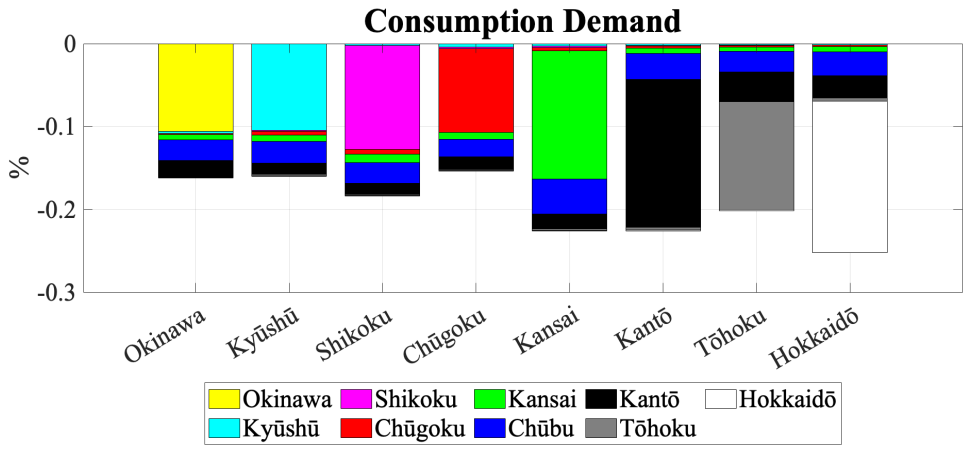


Figure 23: Changes in consumption with 2009Q1's shock to the TE in Chūbu (2008Q3=1).

demand component is largely driven by a decline in Chūbu's demand for that region. The decline in a region's demand for its own goods and services also contributes to the overall decline, which is particularly important for explaining the consumption decline.

(B) The role of price flexibility

Once again, we consider a situation where all prices are fixed at the level of $t = 0$.

Figures 25 to 27 present results comparable to those for the flexible price benchmark. Now, the outputs of all other regions move negatively. The absence of a price effect implies that consumption demand and intermediate-good demand directly affect the output of other regions.

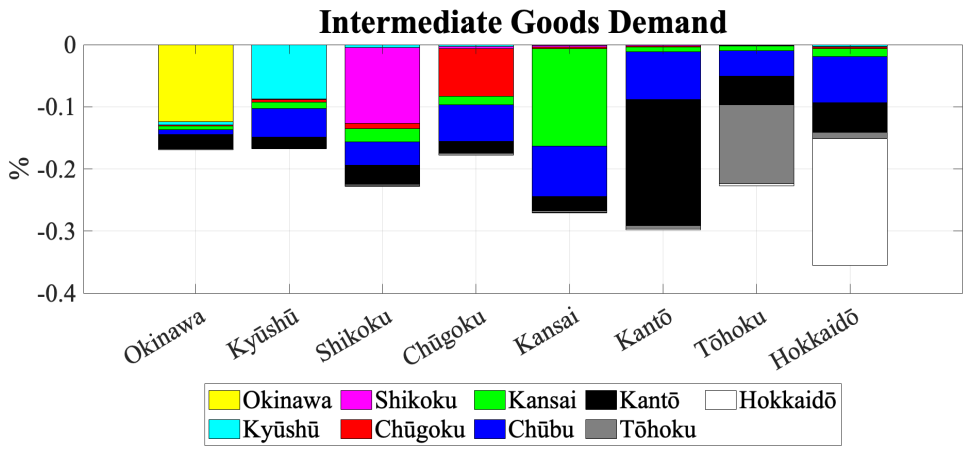


Figure 24: Changes in intermediate goods demand with 2009Q1's shock to the TE in Chūbu (2008Q3=1).

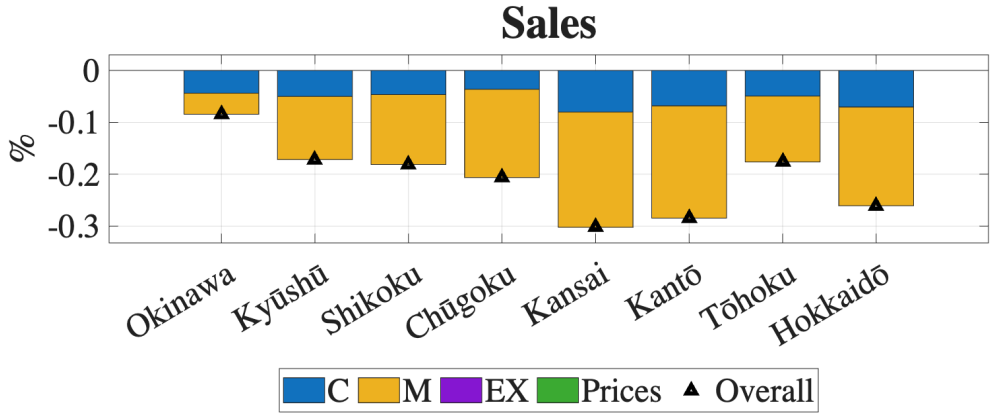


Figure 25: Changes in output with 2009Q1's shock to the TE in Chūbu (2008Q3=1, fixed prices).

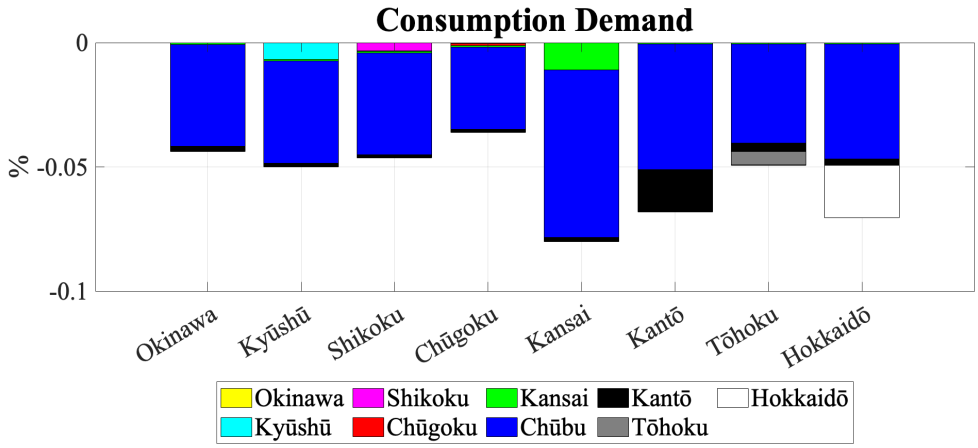


Figure 26: Changes in consumption with 2009Q1's shock to the TE in Chūbu (2008Q3=1, fixed prices).

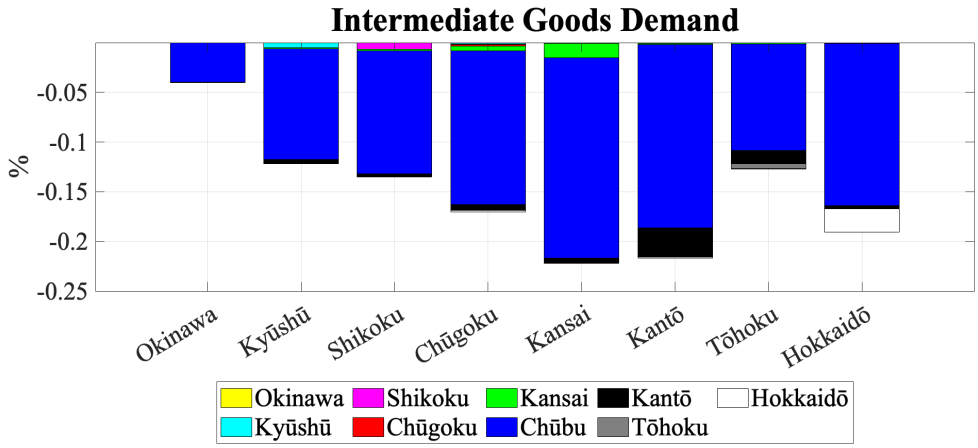


Figure 27: Changes in intermediate goods demand with 2009Q1's shock to the TE in Chūbu (2008Q3=1, fixed prices).

G Computation of the dynamic model

This section details the computation of our main quantitative model.

Steady state: The steady state of the model economy is computed by the following steps.

1. Normalize the foreign price level $\bar{P} = 1$. Normalize the import prices $p_{sf} = 1$.

2. Guess w_{si} for all (s, i), p_{si} for all (s, i), and r_i for all i. Then compute the price indices P_i^c and P_{si}^m from (5) and (9). Compute the unit cost λ_{si} from (14), and then verify p_{si} using the markup formula (16). Thus, for a given (w_{si}, r_i), this routine yields p_{si} consistent with that (w_{si}, r_i).

3. Using the computed prices p_{si}, define the price indices for investment goods P_i^x for each i. Given the guessed nominal interest rates r_i, check whether the implied real interest rate r_i/P_i^x equals $\rho + \delta$ for each i; if not, update r_i and return to step 2.

4. Further guess D_{si} for all (s, i). Then compute y_{sj}, M_{sj}, N_{sj}, and K_i using (17), (18), (19), and (20). Compute Π_i from (22). Then compute the expenditure $E_i = \sum_{s=1}^{S} w_{si} n_{si} + r_i K_i + \Pi_i + B^i$. In the steady state, $X_i = \delta K_i$ must hold. Then use the budget constraint (2) to compute C_i, and use M_{sj} and C_i in (13) to check whether the initial guess on D_{si} was correct.

5. Finally, check w_{si}. w_{si} can be checked using (7). c_{sf}^i is given by (4), which is computed using P_i^c from step 2 and C_i from step 4.

Transition dynamics: Consider the time path of new export parameter values from $t = 1$ (i.e., 2008Q4 in our exercise) onward, with the economy reaching the final steady state at $t = T$. The final period T is set to 2010Q4.

1. Compute the final steady state at period T.

2. Guess sequences of nominal interest rates for each region, $\{\tilde{r}_{i,t}\}_{i,t=1,\ldots,T-1}$.

3. Given $\{\tilde{r}_{i,t}\}_{i,t=1,\ldots,T-1}$ and $\{K_{i,T}\}_i$, implement the following subroutine for each t, working backward from $t = T - 1$. The algorithm of this subroutine is the same as that used to solve the steady state, except for two points: (1) we do not update the guess for nominal interest rates in this subroutine, and (2) the investment $X_{i,t}$ is given by $K_{i,t+1} - (1-\delta)K_{i,t}$, not $\delta K_{i,t}$.

    - Normalize the foreign price level $\bar{P} = 1$ and the import prices $p_{sf} = 1$.
    - Guess w_{si} for all (s, i) and p_{si} for all (s, i). Then compute the price indices P_i^c and P_{si}^m from (5) and (9). The unit cost λ_{si} can be computed from (14), and p_{si} can be verified using the markup formula (16). Thus, for a given (w_{si}, r_i), this routine yields p_{si} consistent with that (w_{si}, r_i).
    - Further guess D_{si} for all (s, i). Then compute y_{sj}, M_{sj}, N_{sj}, and K_i using (17), (18), (19), and (20). Compute Π_i from (22). Then compute $E_i = \sum_{s=1}^{S} w_{si} n_{si} + r_i K_i + \Pi_i + B^i$. $X_{i,t}$ is given by $K_{i,t+1} - (1-\delta)K_{i,t}$. Then use the budget constraint (2) to compute C_i, and use M_{sj} and C_i in (13) to check whether the initial guess on D_{si} was correct.

- Finally, check w_{si}. w_{si} can be checked using (7). c^i_{sf} is given by (4), computed using P^c_i from step 2 and C_i from step 4.

4. Check whether the implied allocations satisfy the following conditions for each t. If not, update the guesses for $r_{i,t}$ and return to step 3:

   - $t = 1, \ldots, T-1$: Check whether the Euler equation (6) is satisfied.
   - $t = 1$: Check whether capital markets clear.
     - The capital supply is fixed at its initial steady-state level for each region.
     - The demand is determined by producers' first-order conditions given the nominal interest rates.